

Assignment1: CancerMortality

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Introduction

This report documents the process and results of the Modeling Assignment for the Statistical Modeling and Inference course. The primary aim of this project is to analyze the Cancer Mortality dataset and build an accurate and interpretable linear model to predict `target_deathrate`, which represents the average cancer mortality rate per 100,000 individuals.

The project begins with data preparation and exploratory analysis, focusing on cleaning and enhancing the dataset to ensure its quality. This involves generating a Data Quality Report, identifying and handling missing values through imputation, and analyzing variable distributions and relationships. Univariate descriptive analysis is performed to examine each variable's characteristics.

The second stage involves modeling using the training dataset, where several key analyses are conducted to optimize the model:

- Initial model exploration addresses potential issues such as multicollinearity and the contribution of each predictor.
- Model transformations are applied to improve variable relationships and address non-linearity.
- An influential data study identifies and evaluates the impact of unusual observations on the model.
- The model is further refined through validation and diagnostics, ensuring it adheres to key assumptions like normality, homoscedasticity, and independence.

Finally, the evaluation process assesses the model's generalization performance on the test dataset, providing insights into its predictive capabilities. A detailed interpretation of the model results is presented, highlighting how the predictors influence the target variable and their practical implications for understanding cancer mortality trends.

This report follows a systematic and methodical approach to ensure the development of a reliable model.

First Steps

Load the dataset and required libraries

```
rm(list=ls())
knitr::opts_chunk$set(fig.align='center', out.width='50%') # Manage output plots width and align

train <- read.csv("train.csv")
test  <- read.csv("test.csv")

library(moments)
library(car)
```

```
## Loading required package: carData
```

```
library(chemometrics)
```

```
## Loading required package: rpart
```

```
library(mice)
```

```
##
```

```
## Attaching package: 'mice'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      filter
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      cbind, rbind
```

```
library(missMDA)
```

```
library(FactoMineR)
```

```
library(corrplot)
```

```
## corrplot 0.94 loaded
```

```
library(VIM)
```

```
## Loading required package: colorspace
```

```
## Loading required package: grid
```

```
## VIM is ready to use.
```

```
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
```

```
##
## Attaching package: 'VIM'

## The following object is masked from 'package:datasets':
##
##     sleep
```

Define some functions to avoid repetition of same processes

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  boxplot(variable, main = "Boxplot")
  outliers <- boxplot.stats(data)$out

  varsumm <- summary(data)
  Q1 <- varsumm[2]
  Q3 <- varsumm[5]
  IQR <- Q3 - Q1

  lower_severe_bound <- Q1 - 3 * IQR
  upper_severe_bound <- Q3 + 3 * IQR
  lower_mild_bound <- Q1 - 1.5 * IQR
  upper_mild_bound <- Q3 + 1.5 * IQR

  # Identify outliers
  mild_outliers <- which((data < lower_mild_bound) | (data > upper_mild_bound))
  severe_outliers <- which((data < lower_severe_bound) | (data > upper_severe_bound))
  num_mild_outliers <- length(mild_outliers)
  num_severe_outliers <- length(severe_outliers)

  # Return the results as a list
  return(list(
    num_outliers = num_mild_outliers,
    num_severe_outliers = num_severe_outliers,
    outliers_indexes = mild_outliers
  ))
}
```

```
summary_and_normality_check <- function(data) {
  #Basic summary
  print("Summary: ")
  print(summary(data))

  # Plot normality
  print("Mean")
  mm <- mean(data, na.rm = TRUE); print(mm)
  print("Standard deviation")
  ss <- sd(data, na.rm = TRUE); print(ss)
  hist(data, freq = F, 100, main = "Histogram")
  curve(dnorm(x, mean = mm, sd = ss), col="red", lwd = 2, add=T)

  #Compute skewness
```

```

print("Skewness")
print(skewness(data, na.rm = TRUE))

#Number of outliers
out.res <- calculate_outliers(data)
print("Num Outliers:")
print(out.res$num_outliers)
print("Num. severe outliers:")
print(out.res$num_severe_outliers)

#Shapiro test
print("Shapiro test: ")
print(shapiro.test(data))

return(out.res$outliers_indexes)
}

outliers_df <- train
outliers_df[,] <- 0

```

Data Preparation & Exploratory Analysis

In this section, a descriptive analysis is conducted on each variable, a data quality report is generated, and imputation and profiling are addressed.

Univariate Descriptive Analysis (Variable Analysis) & Data quality

- Original numeric variables corresponding to qualitative concepts have to be converted to factors:
- Original numeric variables corresponding to real quantitative concepts are kept as numeric but additional factors should also be created
- Exploratory Data Analysis for each variable (numeric summary and graphic support) of the train sample.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” The difference between the mean (623.2) and the median (175) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (80 and 509, respectively) are much lower than the mean of 623.2 and the maximum value of 38150.0, further suggesting skewness. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 333 outliers, and 257 of these are severe outliers.

In this variable there are 126 values = “1962.667684” that considered as NA . Lastly, the skeweness indicator and Shapiro test rejection of the null hypothesis also confirm the non-normal distribution of the variable.

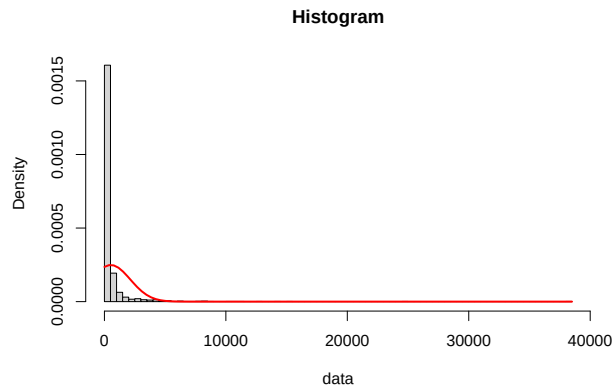
– op 2 – 207 mild / 131 severe

```
variable <- train$avganncount
l11 <- which(variable == 1962.667684 ); length(l11)
```

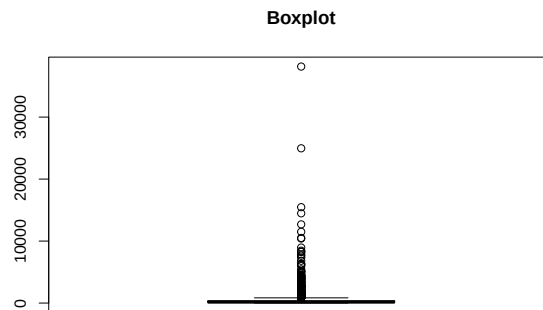
```
## [1] 126
```

```
variable[variable == 1962.667684 ] <- NA
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##       7.0   74.0   155.0   524.2   383.0 38150.0    126
## [1] "Mean"
## [1] 524.2393
## [1] "Standard deviation"
## [1] 1601.779
```



```
## [1] "Skewness"
## [1] 12.16498
```



```
## [1] "Num Outliers:"
## [1] 207
## [1] "Num. severe outliers:"
## [1] 131
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.2677, p-value < 2.2e-16
```

```
outliers_df[res.var,"avganncount"] <- 1
outliers <- which(outliers_df$avganncount == 1); length(outliers)
```

```
## [1] 207
```

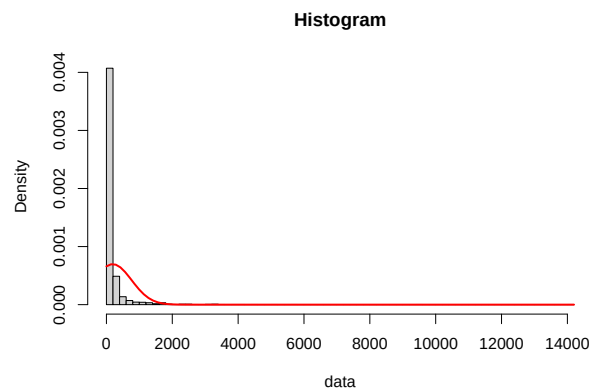
Variable *avgdeathsperyear*

In a similar way, *avgdeathsperyear* is a numerical variable expressing the “Mean number of reported mortalities due to cancer”.

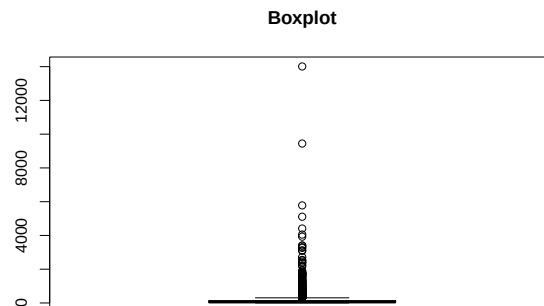
It has no missing values. The difference between the mean (191.6) and median (62) indicates that the distribution is not symmetric. Moreover, the first and third quartiles (29 and 140.5, respectively) are lower than the mean and the maximum value of 14010, further suggesting a skewed ditribution. With the help of graphical tools like the boxplot we can confirm the skeweness, obtaining 225 outliers (138 of which are severe), indicating a large right tail. The histogram overlaid with a normal curve highlights the deviation from a normal distribution with the same mean and standard deviation, and the Shapiro test also confirms the non-normality of the variable.

```
variable <- train$avgdeathsperyear;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.0   29.0   62.0   191.6   140.5  14010.0
## [1] "Mean"
## [1] 191.6237
## [1] "Standard deviation"
## [1] 571.8763
```



```
## [1] "Skewness"
## [1] 12.4999
```



```
## [1] "Num Outliers:"
```

```
## [1] 225
## [1] "Num. severe outliers:"
## [1] 138
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.26769, p-value < 2.2e-16
```

```
outliers_df[res.var,"avgdeathsperyear"] <- 1
```

Variable *target_deathrate* :

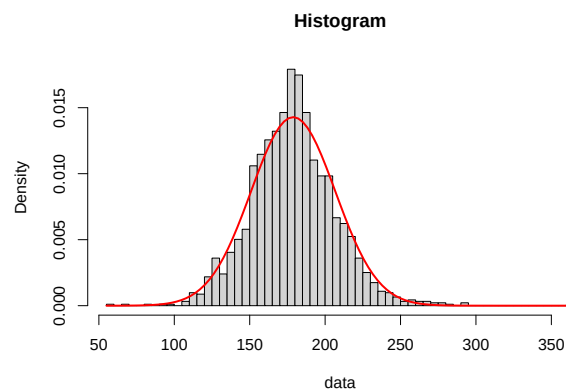
Variable *target_deathrate* is a numerical variable that is the target of the model. It contains no missing values in the train sample. The median (178.3) and mean (178.8) values are really close, expressing the symmetry of the distribution of the variable. Thanks to the boxplot and the histogram we can graphically confirm the symmetry of the distribution.

We performed the Shapiro test to check for the normality of the data, and obtained a small p-value that rejects the null hypothesis, indicating that the variable does not follow a normal distribution. Therefore, the target variable is not normally distributed.

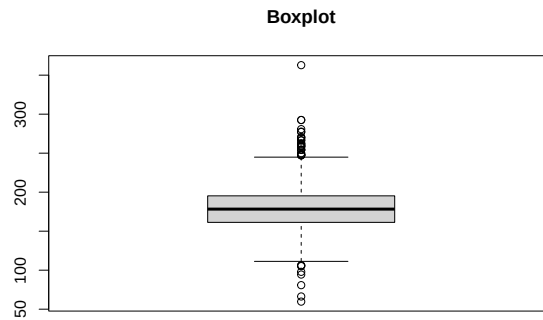
Moreover, it contains 35 outliers, 1 of which is a severe.

```
variable <- train$target_deathrate;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      59.7  161.3   178.3   178.8   195.3   362.8
## [1] "Mean"
## [1] 178.8197
## [1] "Standard deviation"
## [1] 27.93042
```



```
## [1] "Skewness"
## [1] 0.3243216
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 1
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98647, p-value = 4.149e-12
```

```
outliers_df[res.var,"target_deathrate"] <- 1
```

Variable *incidencerate* :

Variable *incidencerate* is a numerical variable that expresses “Mean per capita (100,000) cancer diagnoses”.

The variable has no missing values. It has similar mean (449) and median (453.5) values given its range and suggesting a symmetry distribution. It contains 60 outliers, 7 of which are severe outliers. In the boxplot we can see how the most severe outliers are high and not low values, corresponding to Osceola County, Florida and Williamsburg city, Virginia. The histogram with the overlapping normal distribution shows a quite symmetrical distribution, but the Shapiro test and skeweness value discard the normality of the variable’s distribution.

In this variable there are 126 values = “453.5494221” we should consider these as NA (same rows as avganncount variable)

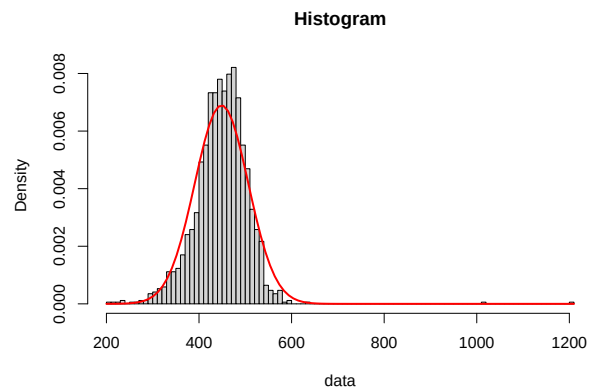
op2 – 36 outliers / 4 severe

```
variable <- train$incidencerate;
l11 <- which(variable == 453.5494221 ); length(l11)
```

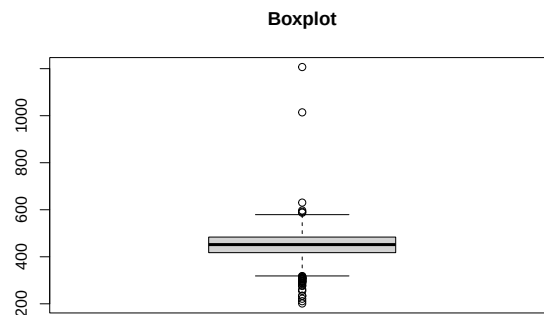
```
## [1] 126
```

```
variable[variable == 453.5494221 ] <- NA
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##    201.3  417.5   451.8   448.7   483.8  1206.9     126
## [1] "Mean"
## [1] 448.6551
## [1] "Standard deviation"
## [1] 57.85323
```



```
## [1] "Skewness"
## [1] 1.397679
```



```
## [1] "Num Outliers:"
## [1] 36
## [1] "Num. severe outliers:"
## [1] 4
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.90204, p-value < 2.2e-16
```

```
outliers_df[res.var,"incidencerate"] <- 1
```

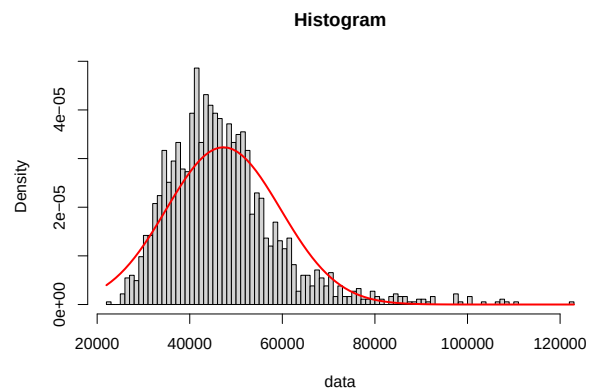
Variable *medincome* :

Variable *medincome* is a numerical variable that expresses the “Median income per county”.

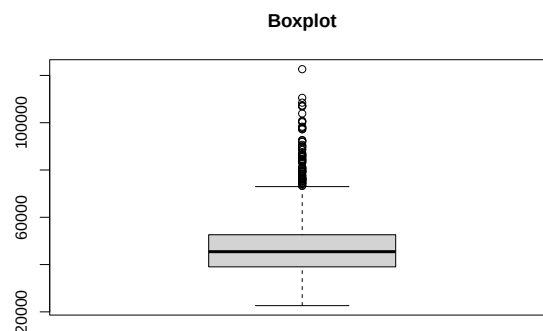
Thanks to the histogram, Shapiro test, and skeweness value, we see how the distribution is neither Normal nor symmetric, but it is skewed with a long right tail. It has 69 outliers, 15 of them severe. The highest value is for Loudoun County, Virginia. There are no missing values in this variable.

```
variable <- train$medincome;  
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
##  22640  39031   45454   47278   52612   122641  
## [1] "Mean"  
## [1] 47277.94  
## [1] "Standard deviation"  
## [1] 12341.69
```



```
## [1] "Skewness"  
## [1] 1.466811
```



```
## [1] "Num Outliers:"
## [1] 69
## [1] "Num. severe outliers:"
## [1] 15
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9105, p-value < 2.2e-16
```

```
outliers_df[res.var,"medincome"] <- 1
```

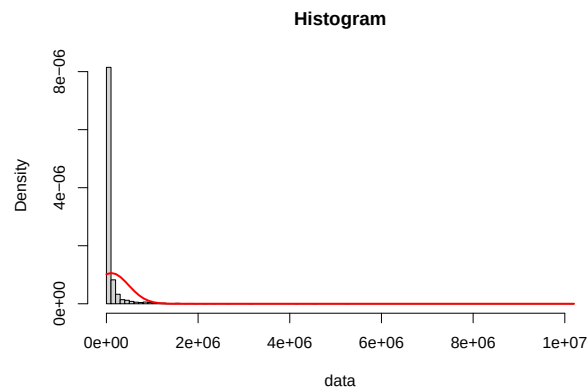
Variable *popest2015* :

Variable *popest2015* is a numerical variable that expresses the population of the county.

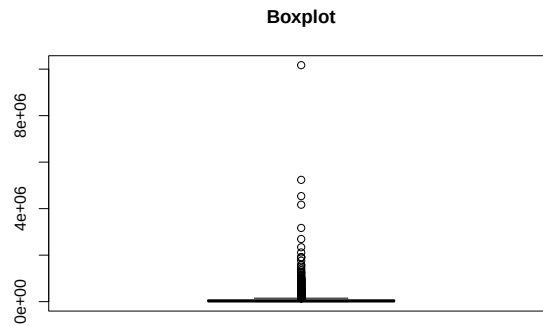
It does not count with missing values. It has a long range and is highly skewed, with a long right tail. In fact, it contains 252 outliers, 162 of which are severe, and all of them are above the upper whisker in the boxplot. The highest value is for Los Angeles County, California. The large difference between the mean (106.841) and median (27.158), along with the spread between the 1st (12.191) and 3rd quartiles (66.880), indicates a positively skewed distribution, where extreme high values are pulling the mean upward.

```
variable <- train$popest2015;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min.  1st Qu.   Median     Mean  3rd Qu.    Max.
##      829   12191   27158   106841   66880 10170292
## [1] "Mean"
## [1] 106841.2
## [1] "Standard deviation"
## [1] 376494
```



```
## [1] "Skewness"
## [1] 14.70818
```



```
## [1] "Num Outliers:"
## [1] 252
## [1] "Num. severe outliers:"
## [1] 162
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.22666, p-value < 2.2e-16
```

```
outliers_df[res.var,"popest2015"] <- 1
```

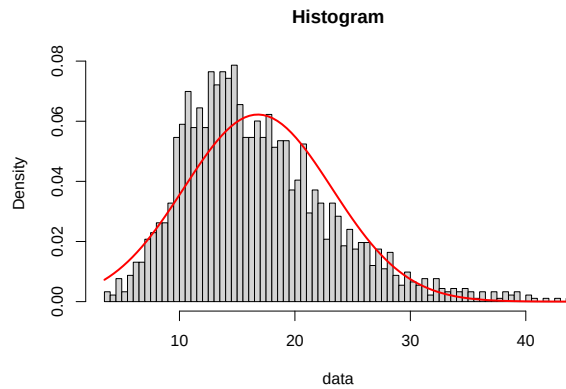
Variable *povertypersent* :

Variable *povertypersent* is a numerical variable that expresses the percent of population in poverty and has no missing values.

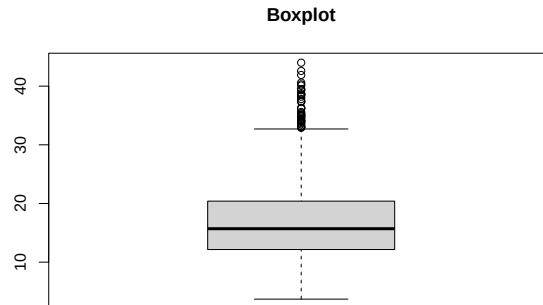
Similarly to other variables, it has a skewed distribution with a larger right tail. It has 42 outliers, but none of them is severe. The p-value obtained in the shapiro_test confirms the rejection of the null hypothesis, concluding that the variable does not have a normal distribution.

```
variable <- train$povertypersent;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.70  12.15   15.70   16.79   20.40   44.00
## [1] "Mean"
## [1] 16.79175
## [1] "Standard deviation"
## [1] 6.411161
```



```
## [1] "Skewness"
## [1] 0.8905168
```



```
## [1] "Num Outliers:"
## [1] 42
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.95557, p-value < 2.2e-16
```

```
outliers_df[res.var,"povertypercent"] <- 1
```

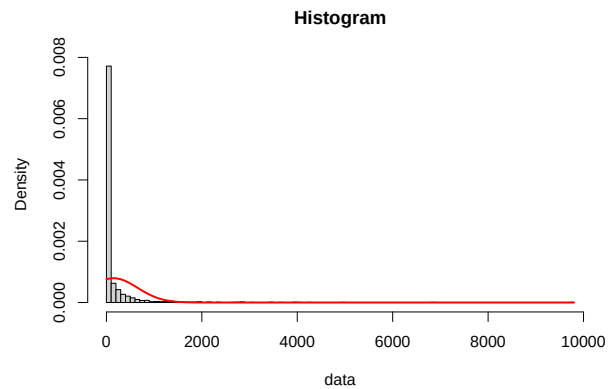
Variable *studypcap* : I CONSIDER IT IS WORTH TO REMOVE

Variable *studypcap* is a numerical variable that expresses “Per capita number of cancer-related clinical trials per county”.

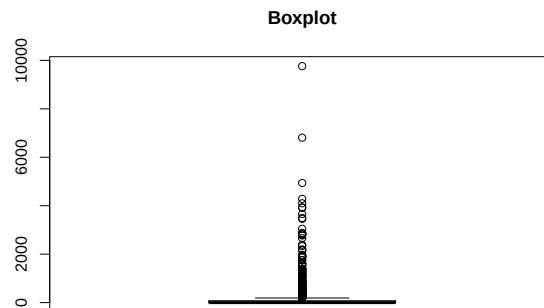
It contains no missing values, and has 307 outliers (222 severe). Its distribution is non-symmetric. Note that 1162 out of 1831 observations contain value 0 in this variable, which indicates that it may only provide noise to the model. **Thus, we consider it is worth to remove the variable.**

```
variable <- train$studypercap;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0     0.0     0.0   148.2   76.0  9762.3
## [1] "Mean"
## [1] 148.2024
## [1] "Standard deviation"
## [1] 501.867
```



```
## [1] "Skewness"
## [1] 8.477802
```



```
## [1] "Num Outliers:"
## [1] 307
## [1] "Num. severe outliers:"
## [1] 222
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.30754, p-value < 2.2e-16
```

```
outliers_df[res.var,"studypercap"] <- 1
```

Variable *binnedi* : I CONSIDER IT IS WORTH TO REMOVE

Variable *binnedi* is a categorical variable (or factor) that expresses the median income per capita binned by decile.

```
variable <- train$binnedinc;
summary <- summary(variable); summary
```

```
##      Length      Class      Mode
##      1831 character character
```

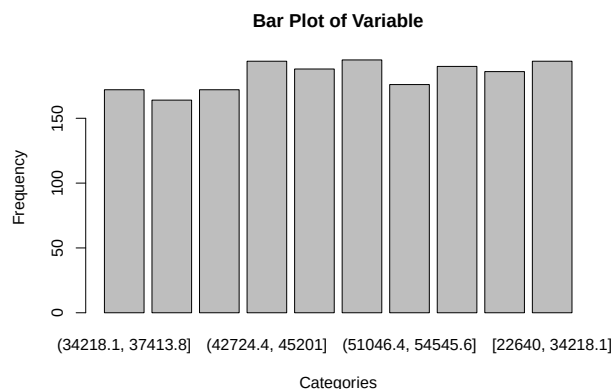
```
table(variable)
```

```
## variable
## (34218.1, 37413.8] (37413.8, 40362.7] (40362.7, 42724.4] (42724.4, 45201]
##              172              164              172              194
## (45201, 48021.6] (48021.6, 51046.4] (51046.4, 54545.6] (54545.6, 61494.5]
##              188              195              176              190
## (61494.5, 125635] [22640, 34218.1]
##              186              194
```

```
prop.table(table(variable))
```

```
## variable
## (34218.1, 37413.8] (37413.8, 40362.7] (40362.7, 42724.4] (42724.4, 45201]
##      0.09393774      0.08956854      0.09393774      0.10595303
## (45201, 48021.6] (48021.6, 51046.4] (51046.4, 54545.6] (54545.6, 61494.5]
##      0.10267613      0.10649918      0.09612234      0.10376843
## (61494.5, 125635] [22640, 34218.1]
##      0.10158383      0.10595303
```

```
barplot(table(variable), main="Bar Plot of Variable", xlab="Categories", ylab="Frequency")
```



Variable *medianage* :

Medianage is a numerical variable that expresses the median age of county residents.

The variable has no missing values. Has 65 outliers, 18 of which are severe outliers. All of these outliers should be removed because it's impossible to be more than 100 years. Once we removed all the incorrect values we get 52 outliers, 18 of them severe. Finally we observe a symmetric distribution with a median = 40.90 and a mean=40.85. After analyzing the boxplot, and the histogram along the Shapiro test we can discard the normality of the variable's distribution.

```
variable <- train$medianage  
table(variable)
```

```
## variable  
## 23.3 24.2 24.4 24.6 25 25.9 26.1 26.3 26.4 26.5 26.6 26.8 27.2  
## 1 1 1 1 1 1 2 1 1 1 1 1 1  
## 27.5 27.8 27.9 28 28.1 28.3 28.4 28.5 28.6 28.8 29.1 29.4 29.8  
## 1 1 2 1 1 2 1 1 1 2 2 3 3  
## 29.9 30 30.1 30.2 30.3 30.4 30.5 30.6 30.7 30.8 30.9 31 31.1  
## 1 3 1 5 4 4 2 2 5 3 2 1 4  
## 31.2 31.4 31.5 31.6 31.7 31.8 31.9 32 32.1 32.2 32.3 32.4 32.5  
## 1 2 4 2 4 2 4 2 2 4 1 4 3  
## 32.6 32.7 32.8 32.9 33 33.1 33.2 33.3 33.4 33.5 33.6 33.7 33.8  
## 1 6 5 4 4 1 2 4 1 6 2 2 5  
## 33.9 34 34.1 34.2 34.3 34.4 34.5 34.6 34.7 34.8 34.9 35 35.1  
## 3 7 7 3 2 3 3 7 7 12 6 5 4  
## 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1 36.2 36.3 36.4  
## 6 7 5 10 11 7 7 5 10 14 6 7 7  
## 36.5 36.6 36.7 36.8 36.9 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7  
## 10 9 10 11 9 10 7 8 8 11 10 13 6  
## 37.8 37.9 38 38.1 38.2 38.3 38.4 38.5 38.6 38.7 38.8 38.9 39  
## 10 12 13 10 9 9 15 17 10 7 29 13 21  
## 39.1 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9 40 40.1 40.2 40.3  
## 17 14 16 15 14 10 18 17 18 14 20 11 16  
## 40.4 40.5 40.6 40.7 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5 41.6  
## 12 14 17 22 22 13 24 17 17 19 16 19 15  
## 41.7 41.8 41.9 42 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9  
## 11 17 17 15 16 19 18 14 14 22 9 15 16  
## 43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9 44 44.1 44.2  
## 14 10 12 10 9 17 13 16 11 10 5 14 14  
## 44.3 44.4 44.5 44.6 44.7 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5  
## 10 10 11 8 8 5 10 8 8 7 5 11 6  
## 45.6 45.7 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8  
## 8 8 8 7 10 8 6 5 8 6 7 6 6  
## 46.9 47 47.1 47.2 47.3 47.4 47.5 47.6 47.7 47.8 47.9 48 48.1  
## 5 4 8 6 4 4 6 7 4 5 5 4 3  
## 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9 49 49.1 49.2 49.3 49.4  
## 4 7 4 3 4 2 1 2 4 8 5 2 5  
## 49.5 49.6 49.7 49.8 49.9 50 50.1 50.2 50.3 50.4 50.5 50.6 50.7  
## 4 4 1 6 3 3 2 3 1 2 2 2 4  
## 50.8 50.9 51 51.1 51.2 51.3 51.4 51.5 51.7 51.8 51.9 52 52.1  
## 3 4 1 2 2 5 1 1 1 2 2 1 2  
## 52.3 52.4 52.5 52.6 52.8 52.9 53.1 53.2 53.3 53.4 53.5 54.1 54.4  
## 2 1 1 1 1 1 1 1 1 1 2 1 2
```

```
## 54.5 54.6 54.7 54.8 55.4 55.6 56.3 56.4 56.5 56.6 59 349.2 406.8
## 1 2 1 1 1 3 1 1 1 1 1 1 1
## 424.8 429.6 430.8 481.2 496.8 498 500.4 502.8 508.8 523.2 525.6 536.4 546
## 1 1 1 1 2 1 1 1 2 1 1 1 1
## 624
## 1
```

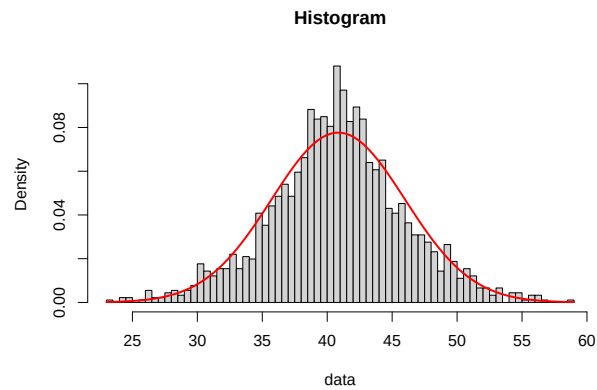
```
variable[variable > 70 ] <- NA
table(variable)
```

```
## variable
## 23.3 24.2 24.4 24.6 25 25.9 26.1 26.3 26.4 26.5 26.6 26.8 27.2 27.5 27.8 27.9
## 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 2
## 28 28.1 28.3 28.4 28.5 28.6 28.8 29.1 29.4 29.8 29.9 30 30.1 30.2 30.3 30.4
## 1 1 2 1 1 1 2 2 3 3 1 3 1 5 4 4
## 30.5 30.6 30.7 30.8 30.9 31 31.1 31.2 31.4 31.5 31.6 31.7 31.8 31.9 32 32.1
## 2 2 5 3 2 1 4 1 2 4 2 4 2 4 2 2
## 32.2 32.3 32.4 32.5 32.6 32.7 32.8 32.9 33 33.1 33.2 33.3 33.4 33.5 33.6 33.7
## 4 1 4 3 1 6 5 4 4 1 2 4 1 6 2 2
## 33.8 33.9 34 34.1 34.2 34.3 34.4 34.5 34.6 34.7 34.8 34.9 35 35.1 35.2 35.3
## 5 3 7 7 3 2 3 3 7 7 12 6 5 4 6 7
## 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1 36.2 36.3 36.4 36.5 36.6 36.7 36.8 36.9
## 5 10 11 7 7 5 10 14 6 7 7 10 9 10 11 9
## 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7 37.8 37.9 38 38.1 38.2 38.3 38.4 38.5
## 10 7 8 8 11 10 13 6 10 12 13 10 9 9 15 17
## 38.6 38.7 38.8 38.9 39 39.1 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9 40 40.1
## 10 7 29 13 21 17 14 16 15 14 10 18 17 18 14 20
## 40.2 40.3 40.4 40.5 40.6 40.7 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5 41.6 41.7
## 11 16 12 14 17 22 22 13 24 17 17 19 16 19 15 11
## 41.8 41.9 42 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9 43 43.1 43.2 43.3
## 17 17 15 16 19 18 14 14 22 9 15 16 14 10 12 10
## 43.4 43.5 43.6 43.7 43.8 43.9 44 44.1 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9
## 9 17 13 16 11 10 5 14 14 10 10 11 8 8 5 10
## 45 45.1 45.2 45.3 45.4 45.5 45.6 45.7 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5
## 8 8 7 5 11 6 8 8 8 7 10 8 6 5 8 6
## 46.6 46.7 46.8 46.9 47 47.1 47.2 47.3 47.4 47.5 47.6 47.7 47.8 47.9 48 48.1
## 7 6 6 5 4 8 6 4 4 6 7 4 5 5 4 3
## 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9 49 49.1 49.2 49.3 49.4 49.5 49.6 49.7
## 4 7 4 3 4 2 1 2 4 8 5 2 5 4 4 1
## 49.8 49.9 50 50.1 50.2 50.3 50.4 50.5 50.6 50.7 50.8 50.9 51 51.1 51.2 51.3
## 6 3 3 2 3 1 2 2 2 4 3 4 1 2 2 5
## 51.4 51.5 51.7 51.8 51.9 52 52.1 52.3 52.4 52.5 52.6 52.8 52.9 53.1 53.2 53.3
## 1 1 1 2 2 1 2 2 1 1 1 1 1 1 1 1
## 53.4 53.5 54.1 54.4 54.5 54.6 54.7 54.8 55.4 55.6 56.3 56.4 56.5 56.6 59
## 1 2 1 2 1 2 1 1 1 3 1 1 1 1 1
```

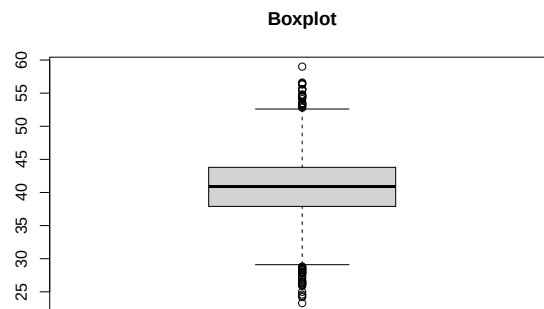
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
## Min. 1st Qu. Median Mean 3rd Qu. Max. NA's
## 23.30 37.90 40.90 40.85 43.80 59.00 18
## [1] "Mean"
## [1] 40.85063
```

```
## [1] "Standard deviation"
## [1] 5.13671
```



```
## [1] "Skewness"
## [1] -0.02432015
```



```
## [1] "Num Outliers:"
## [1] 52
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9951, p-value = 1.161e-05
```

```
outliers_df[res.var,"medianage"] <- 1
```

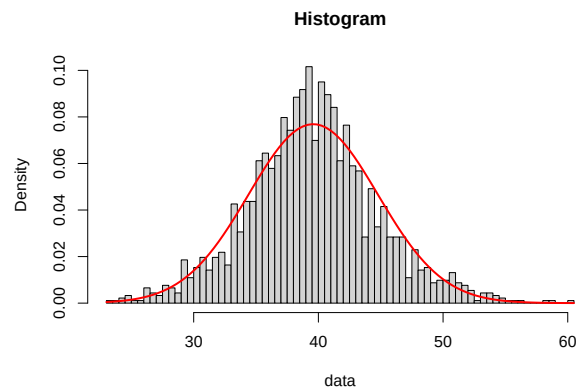
Variable *medianagemale* :

Numeric variable, represents the mean age of the male population of each county.

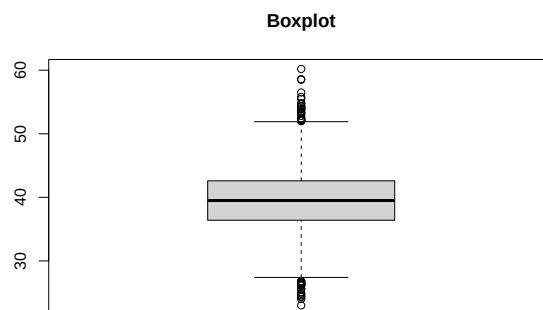
There are no missing values and it has 46 outliers, non of them several outliers. The mean is 39.59 yo and the median 39.50yo what suggest a simmetry distribution. After looking at the results of the shapiro-test it doesn't have a normal distribution

```
variable <- train$medianagemale
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 23.00  36.40   39.50   39.59  42.60   60.20
## [1] "Mean"
## [1] 39.58564
## [1] "Standard deviation"
## [1] 5.188667
```



```
## [1] "Skewness"
## [1] 0.1612965
```



```
## [1] "Num Outliers:"
## [1] 46
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
```

```
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99404, p-value = 9.877e-07
```

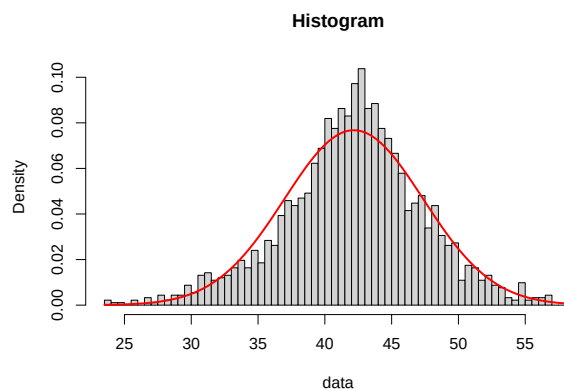
```
outliers_df[res.var, "medianagemale"] <- 1
```

Variable *medianagemale* : I AM NOT SURE IF IT'S WORTH IT TO SEPARATE THE MEDIANAGE WITH SEX

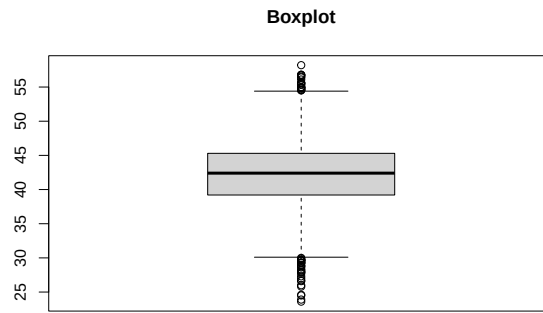
The medianagemale is a numerical variable that analyze the median age of the females in each county. It has a simmetric distribution (median=42.40, mean=42,17) but after analyzing the results of the Shapiro-test we observe that does not have a normal distribution. There are 55 outliers none of them severe

```
variable <- train$medianagemale
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  23.60  39.20   42.40   42.17  45.30   58.20
## [1] "Mean"
## [1] 42.16931
## [1] "Standard deviation"
## [1] 5.196007
```



```
## [1] "Skewness"
## [1] -0.2057034
```



```
## [1] "Num Outliers:"
## [1] 55
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99321, p-value = 1.817e-07
```

```
outliers_df[res.var,"medianagefemale"] <- 1
```

Variable *geography* :

Categorical variable that shows the name of each county. We can use this variable as a PK

–comment – Maybe we can separate the name of the state and create a new variable to be able to study the distribution around the United States

```
variable <- train$geography;
summary <- summary(variable); summary
```

```
##      Length      Class      Mode
##      1831 character character
```

```
table(variable)
```

```
## variable
##          Acadia Parish, Louisiana          Ada County, Idaho
##                      1                      1
##          Adair County, Kentucky          Adair County, Missouri
##                      1                      1
##          Adams County, Idaho          Adams County, Illinois
##                      1                      1
##          Adams County, Iowa          Adams County, Mississippi
##                      1                      1
```

##	Adams County, Nebraska	Adams County, North Dakota
##	1	1
##	Adams County, Ohio	Aiken County, South Carolina
##	1	1
##	Alachua County, Florida	Albany County, Wyoming
##	1	1
##	Albemarle County, Virginia	Alcona County, Michigan
##	1	1
##	Alcorn County, Mississippi	Aleutians West Census Area, Alaska
##	1	1
##	Alexander County, Illinois	Alfalfa County, Oklahoma
##	1	1
##	Alger County, Michigan	Allegan County, Michigan
##	1	1
##	Allegany County, New York	Alleghany County, Virginia
##	1	1
##	Allen County, Kansas	Allen Parish, Louisiana
##	1	1
##	Alpena County, Michigan	Amador County, California
##	1	1
##	Amelia County, Virginia	Amite County, Mississippi
##	1	1
##	Anchorage Municipality, Alaska	Anderson County, Kentucky
##	1	1
##	Anderson County, South Carolina	Andrew County, Missouri
##	1	1
##	Androscoggin County, Maine	Anne Arundel County, Maryland
##	1	1
##	Anoka County, Minnesota	Anson County, North Carolina
##	1	1
##	Antelope County, Nebraska	Antrim County, Michigan
##	1	1
##	Appanoose County, Iowa	Appling County, Georgia
##	1	1
##	Arapahoe County, Colorado	Archer County, Texas
##	1	1
##	Archuleta County, Colorado	Arenac County, Michigan
##	1	1
##	Arlington County, Virginia	Armstrong County, Pennsylvania
##	1	1
##	Armstrong County, Texas	Aroostook County, Maine
##	1	1
##	Ascension Parish, Louisiana	Ashe County, North Carolina
##	1	1
##	Ashland County, Ohio	Ashtabula County, Ohio
##	1	1
##	Assumption Parish, Louisiana	Atchison County, Kansas
##	1	1
##	Atchison County, Missouri	Athens County, Ohio
##	1	1
##	Atlantic County, New Jersey	Atoka County, Oklahoma
##	1	1
##	Attala County, Mississippi	Audubon County, Iowa
##	1	1

##	Augusta County, Virginia	Aurora County, South Dakota
##	1	1
##	Austin County, Texas	Avoyelles Parish, Louisiana
##	1	1
##	Baca County, Colorado	Bacon County, Georgia
##	1	1
##	Baker County, Florida	Baker County, Georgia
##	1	1
##	Baker County, Oregon	Baltimore city, Maryland
##	1	1
##	Baltimore County, Maryland	Bamberg County, South Carolina
##	1	1
##	Banks County, Georgia	Bannock County, Idaho
##	1	1
##	Barbour County, Alabama	Barbour County, West Virginia
##	1	1
##	Barnes County, North Dakota	Barren County, Kentucky
##	1	1
##	Barron County, Wisconsin	Barrow County, Georgia
##	1	1
##	Barry County, Michigan	Barton County, Missouri
##	1	1
##	Bastrop County, Texas	Bates County, Missouri
##	1	1
##	Bath County, Virginia	Bayfield County, Wisconsin
##	1	1
##	Beadle County, South Dakota	Bear Lake County, Idaho
##	1	1
##	Beaufort County, North Carolina	Beaufort County, South Carolina
##	1	1
##	Beaver County, Oklahoma	Beaver County, Utah
##	1	1
##	Beaverhead County, Montana	Bedford County, Tennessee
##	1	1
##	Bee County, Texas	Belknap County, New Hampshire
##	1	1
##	Bell County, Texas	Beltrami County, Minnesota
##	1	1
##	Benewah County, Idaho	Bennington County, Vermont
##	1	1
##	Benson County, North Dakota	Benton County, Indiana
##	1	1
##	Benton County, Minnesota	Benton County, Mississippi
##	1	1
##	Benton County, Missouri	Benton County, Oregon
##	1	1
##	Benton County, Tennessee	Benton County, Washington
##	1	1
##	Berkeley County, South Carolina	Berkeley County, West Virginia
##	1	1
##	Berkshire County, Massachusetts	Bernalillo County, New Mexico
##	1	1
##	Berrien County, Georgia	Berrien County, Michigan
##	1	1

## Bertie County, North Carolina	1	Bexar County, Texas	1
## Bibb County, Georgia	1	Bienville Parish, Louisiana	1
## Big Horn County, Montana	1	Big Horn County, Wyoming	1
## Black Hawk County, Iowa	1	Blaine County, Idaho	1
## Blaine County, Oklahoma	1	Blair County, Pennsylvania	1
## Blanco County, Texas	1	Blue Earth County, Minnesota	1
## Bolivar County, Mississippi	1	Bon Homme County, South Dakota	1
## Bonner County, Idaho	1	Boone County, Arkansas	1
## Boone County, Illinois	1	Boone County, Indiana	1
## Bosque County, Texas	1	Botetourt County, Virginia	1
## Bottineau County, North Dakota	1	Bourbon County, Kansas	1
## Bowman County, North Dakota	1	Box Elder County, Utah	1
## Boyd County, Kentucky	1	Boyle County, Kentucky	1
## Bradford County, Pennsylvania	1	Bradley County, Arkansas	1
## Bradley County, Tennessee	1	Brantley County, Georgia	1
## Braxton County, West Virginia	1	Breathitt County, Kentucky	1
## Breckinridge County, Kentucky	1	Bremer County, Iowa	1
## Brevard County, Florida	1	Bristol County, Rhode Island	1
## Broadwater County, Montana	1	Brookings County, South Dakota	1
## Brooks County, Georgia	1	Brooks County, Texas	1
## Broome County, New York	1	Broomfield County, Colorado	1
## Broward County, Florida	1	Brown County, Kansas	1
## Brown County, Ohio	1	Brown County, Texas	1
## Brule County, South Dakota	1	Brunswick County, North Carolina	1
## Brunswick County, Virginia	1	Bryan County, Oklahoma	1
## Buchanan County, Iowa	1	Buchanan County, Missouri	1
## Buchanan County, Virginia	1	Buckingham County, Virginia	1

##	Buena Vista County, Iowa	1	Buffalo County, Wisconsin	1
##	Bullitt County, Kentucky	1	Bulloch County, Georgia	1
##	Bullock County, Alabama	1	Buncombe County, North Carolina	1
##	Burke County, Georgia	1	Burke County, North Dakota	1
##	Burleson County, Texas	1	Burlington County, New Jersey	1
##	Burnet County, Texas	1	Burnett County, Wisconsin	1
##	Burt County, Nebraska	1	Butler County, Ohio	1
##	Butte County, South Dakota	1	Cabarrus County, North Carolina	1
##	Caddo County, Oklahoma	1	Calaveras County, California	1
##	Calcasieu Parish, Louisiana	1	Caldwell County, Kentucky	1
##	Caldwell County, Missouri	1	Caldwell County, North Carolina	1
##	Caldwell County, Texas	1	Caldwell Parish, Louisiana	1
##	Calhoun County, Arkansas	1	Calhoun County, Georgia	1
##	Calhoun County, Iowa	1	Calhoun County, Michigan	1
##	Calhoun County, South Carolina	1	Calhoun County, West Virginia	1
##	Callahan County, Texas	1	Calumet County, Wisconsin	1
##	Camden County, Missouri	1	Camden County, North Carolina	1
##	Cameron County, Pennsylvania	1	Cameron Parish, Louisiana	1
##	Campbell County, Kentucky	1	Campbell County, Virginia	1
##	Campbell County, Wyoming	1	Canadian County, Oklahoma	1
##	Cannon County, Tennessee	1	Cape Girardeau County, Missouri	1
##	Cape May County, New Jersey	1	Carbon County, Montana	1
##	Carbon County, Utah	1	Carbon County, Wyoming	1
##	Carlisle County, Kentucky	1	Carlton County, Minnesota	1
##	Caroline County, Virginia	1	Carroll County, Georgia	1
##	Carroll County, Illinois	1	Carroll County, Indiana	1
##	Carroll County, Iowa	1	Carroll County, Kentucky	1

##	Carroll County, Maryland	Carroll County, New Hampshire
##	1	1
##	Carroll County, Ohio	Carroll County, Tennessee
##	1	1
##	Carroll County, Virginia	Carson City, Nevada
##	1	1
##	Carson County, Texas	Carter County, Kentucky
##	1	1
##	Carter County, Tennessee	Carver County, Minnesota
##	1	1
##	Casey County, Kentucky	Cass County, Iowa
##	1	1
##	Cass County, Michigan	Cass County, Missouri
##	1	1
##	Cass County, Nebraska	Cass County, North Dakota
##	1	1
##	Cass County, Texas	Cassia County, Idaho
##	1	1
##	Castro County, Texas	Caswell County, North Carolina
##	1	1
##	Catahoula Parish, Louisiana	Catawba County, North Carolina
##	1	1
##	Catoosa County, Georgia	Cattaraugus County, New York
##	1	1
##	Cavalier County, North Dakota	Cayuga County, New York
##	1	1
##	Cedar County, Iowa	Cedar County, Missouri
##	1	1
##	Cedar County, Nebraska	Centre County, Pennsylvania
##	1	1
##	Cerro Gordo County, Iowa	Chaffee County, Colorado
##	1	1
##	Chambers County, Alabama	Chambers County, Texas
##	1	1
##	Champaign County, Ohio	Chariton County, Missouri
##	1	1
##	Charles Mix County, South Dakota	Charleston County, South Carolina
##	1	1
##	Charlotte County, Virginia	Charlton County, Georgia
##	1	1
##	Chase County, Kansas	Chatham County, Georgia
##	1	1
##	Chattahoochee County, Georgia	Chattooga County, Georgia
##	1	1
##	Cheatham County, Tennessee	Cheboygan County, Michigan
##	1	1
##	Chelan County, Washington	Chemung County, New York
##	1	1
##	Chenango County, New York	Cherokee County, Iowa
##	1	1
##	Cherokee County, North Carolina	Cherokee County, Oklahoma
##	1	1
##	Cherokee County, South Carolina	Chester County, Pennsylvania
##	1	1

##	Chester County, South Carolina	Chester County, Tennessee
##	1	1
##	Chesterfield County, South Carolina	Cheyenne County, Colorado
##	1	1
##	Chickasaw County, Iowa	Chickasaw County, Mississippi
##	1	1
##	Childress County, Texas	Chilton County, Alabama
##	1	1
##	Chippewa County, Michigan	Chippewa County, Minnesota
##	1	1
##	Chisago County, Minnesota	Chittenden County, Vermont
##	1	1
##	Choctaw County, Oklahoma	Chouteau County, Montana
##	1	1
##	Churchill County, Nevada	Cibola County, New Mexico
##	1	1
##	Cimarron County, Oklahoma	Citrus County, Florida
##	1	1
##	Claiborne County, Mississippi	Claiborne County, Tennessee
##	1	1
##	Clallam County, Washington	Clare County, Michigan
##	1	1
##	Clarendon County, South Carolina	Clark County, Missouri
##	1	1
##	Clark County, Nevada	Clark County, South Dakota
##	1	1
##	Clark County, Washington	Clark County, Wisconsin
##	1	1
##	Clarke County, Georgia	Clarke County, Mississippi
##	1	1
##	Clay County, Florida	Clay County, Georgia
##	1	1
##	Clay County, Illinois	Clay County, Iowa
##	1	1
##	Clay County, Kansas	Clay County, Kentucky
##	1	1
##	Clay County, Minnesota	Clay County, Mississippi
##	1	1
##	Clay County, Nebraska	Clay County, North Carolina
##	1	1
##	Clay County, Tennessee	Clay County, Texas
##	1	1
##	Clay County, West Virginia	Clayton County, Georgia
##	1	1
##	Clayton County, Iowa	Clearfield County, Pennsylvania
##	1	1
##	Cleburne County, Alabama	Cleburne County, Arkansas
##	1	1
##	Cleveland County, North Carolina	Clinch County, Georgia
##	1	1
##	Clinton County, Iowa	Clinton County, Michigan
##	1	1
##	Clinton County, New York	Clinton County, Pennsylvania
##	1	1

##	Cloud County, Kansas	Coahoma County, Mississippi
##	1	1
##	Cochise County, Arizona	Cochran County, Texas
##	1	1
##	Cocke County, Tennessee	Coconino County, Arizona
##	1	1
##	Coffee County, Alabama	Coffee County, Georgia
##	1	1
##	Coffee County, Tennessee	Coffey County, Kansas
##	1	1
##	Colbert County, Alabama	Colfax County, Nebraska
##	1	1
##	Colfax County, New Mexico	Collier County, Florida
##	1	1
##	Colonial Heights city, Virginia	Colquitt County, Georgia
##	1	1
##	Columbia County, Georgia	Columbia County, New York
##	1	1
##	Columbia County, Oregon	Columbia County, Washington
##	1	1
##	Columbia County, Wisconsin	Columbiana County, Ohio
##	1	1
##	Columbus County, North Carolina	Colusa County, California
##	1	1
##	Comal County, Texas	Concho County, Texas
##	1	1
##	Concordia Parish, Louisiana	Contra Costa County, California
##	1	1
##	Conway County, Arkansas	Cook County, Georgia
##	1	1
##	Cook County, Illinois	Cook County, Minnesota
##	1	1
##	Cooke County, Texas	Cooper County, Missouri
##	1	1
##	Coos County, New Hampshire	Coosa County, Alabama
##	1	1
##	Corson County, South Dakota	Cortland County, New York
##	1	1
##	Coryell County, Texas	Cottle County, Texas
##	1	1
##	Covington County, Mississippi	Cowley County, Kansas
##	1	1
##	Cowlitz County, Washington	Crane County, Texas
##	1	1
##	Craven County, North Carolina	Crawford County, Arkansas
##	1	1
##	Crawford County, Georgia	Crawford County, Indiana
##	1	1
##	Crawford County, Iowa	Crawford County, Michigan
##	1	1
##	Crawford County, Ohio	Crawford County, Wisconsin
##	1	1
##	Crenshaw County, Alabama	Crisp County, Georgia
##	1	1

##	Crittenden County, Arkansas	Crittenden County, Kentucky
##	1	1
##	Crockett County, Tennessee	Crockett County, Texas
##	1	1
##	Cross County, Arkansas	Crow Wing County, Minnesota
##	1	1
##	Cullman County, Alabama	Culpeper County, Virginia
##	1	1
##	Cumberland County, Kentucky	Cumberland County, Maine
##	1	1
##	Cumberland County, Tennessee	Cuming County, Nebraska
##	1	1
##	Curry County, New Mexico	Curry County, Oregon
##	1	1
##	Custer County, Colorado	Custer County, Oklahoma
##	1	1
##	Cuyahoga County, Ohio	Dade County, Georgia
##	1	1
##	Dade County, Missouri	Dakota County, Minnesota
##	1	1
##	Dale County, Alabama	Dallam County, Texas
##	1	1
##	Dallas County, Alabama	Dallas County, Iowa
##	1	1
##	Dallas County, Missouri	Dane County, Wisconsin
##	1	1
##	Danville city, Virginia	Dare County, North Carolina
##	1	1
##	Darlington County, South Carolina	Davidson County, North Carolina
##	1	1
##	Daviess County, Indiana	Daviess County, Kentucky
##	1	1
##	Daviess County, Missouri	Davis County, Utah
##	1	1
##	Davison County, South Dakota	Dawes County, Nebraska
##	1	1
##	Dawson County, Montana	Dawson County, Nebraska
##	1	1
##	Dawson County, Texas	Day County, South Dakota
##	1	1
##	De Baca County, New Mexico	De Soto Parish, Louisiana
##	1	1
##	Deaf Smith County, Texas	Dearborn County, Indiana
##	1	1
##	Decatur County, Indiana	Decatur County, Iowa
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##	Decatur County, Kansas	Decatur County, Tennessee
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##	Defiance County, Ohio	DeKalb County, Alabama
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##	DeKalb County, Georgia	DeKalb County, Illinois
##	1	1
##	DeKalb County, Indiana	DeKalb County, Missouri
##	1	1

##	Del Norte County, California	Delaware County, Iowa
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##	Delaware County, Ohio	Delaware County, Oklahoma
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##	Delaware County, Pennsylvania	Delta County, Michigan
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##	Delta County, Texas	Dent County, Missouri
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##	Denver County, Colorado	Des Moines County, Iowa
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##	Deschutes County, Oregon	Desha County, Arkansas
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##	Deuel County, South Dakota	Dewey County, Oklahoma
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##	Dewey County, South Dakota	Dickenson County, Virginia
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##	Dickey County, North Dakota	Dickinson County, Michigan
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##	Dickson County, Tennessee	Dillon County, South Carolina
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##	Dinwiddie County, Virginia	Dixie County, Florida
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##	Dixon County, Nebraska	Doddridge County, West Virginia
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##	Dodge County, Minnesota	Dodge County, Nebraska
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##	Dodge County, Wisconsin	Dolores County, Colorado
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##	Doniphan County, Kansas	Donley County, Texas
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##	Doña Ana County, New Mexico	Dooly County, Georgia
##	1	1
##	Dorchester County, South Carolina	Dougherty County, Georgia
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##	Douglas County, Colorado	Douglas County, Illinois
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##	Douglas County, Kansas	Douglas County, Minnesota
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##	Douglas County, Missouri	Douglas County, Nebraska
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##	Douglas County, Nevada	Douglas County, Washington
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##	Drew County, Arkansas	Dubois County, Indiana
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##	Dubuque County, Iowa	Duchesne County, Utah
##	1	1
##	Dukes County, Massachusetts	Dundy County, Nebraska
##	1	1
##	Dunklin County, Missouri	Dunn County, North Dakota
##	1	1
##	Dunn County, Wisconsin	Duplin County, North Carolina
##	1	1
##	Dutchess County, New York	Duval County, Florida
##	1	1

##	Duval County, Texas	1	Dyer County, Tennessee	1
##	Eagle County, Colorado	1	Early County, Georgia	1
##	East Baton Rouge Parish, Louisiana	1	East Feliciana Parish, Louisiana	1
##	Eastland County, Texas	1	Echols County, Georgia	1
##	Ector County, Texas	1	Eddy County, North Dakota	1
##	Edwards County, Kansas	1	Effingham County, Georgia	1
##	El Dorado County, California	1	El Paso County, Texas	1
##	Elbert County, Georgia	1	Elk County, Pennsylvania	1
##	Elkhart County, Indiana	1	Elko County, Nevada	1
##	Elliott County, Kentucky	1	Elmore County, Alabama	1
##	Elmore County, Idaho	1	Emanuel County, Georgia	1
##	Emery County, Utah	1	Emmet County, Michigan	1
##	Emmons County, North Dakota	1	Erath County, Texas	1
##	Erie County, New York	1	Erie County, Pennsylvania	1
##	Escambia County, Florida	1	Esmeralda County, Nevada	1
##	Essex County, Massachusetts	1	Essex County, New York	1
##	Etowah County, Alabama	1	Eureka County, Nevada	1
##	Fairbanks North Star Borough, Alaska	1	Fairfax County, Virginia	1
##	Fairfield County, Connecticut	1	Fairfield County, Ohio	1
##	Fairfield County, South Carolina	1	Fallon County, Montana	1
##	Fannin County, Georgia	1	Fannin County, Texas	1
##	Faribault County, Minnesota	1	Faulk County, South Dakota	1
##	Faulkner County, Arkansas	1	Fauquier County, Virginia	1
##	Fayette County, Indiana	1	Fayette County, Kentucky	1
##	Fayette County, Ohio	1	Fayette County, Pennsylvania	1
##	Fayette County, Tennessee	1	Fentress County, Tennessee	1
##	Finney County, Kansas	1	Fisher County, Texas	1

##	Flagler County, Florida	Fleming County, Kentucky
##	1	1
##	Florence County, Wisconsin	Floyd County, Indiana
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##	Floyd County, Kentucky	Floyd County, Virginia
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##	Fluvanna County, Virginia	Ford County, Illinois
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##	Ford County, Kansas	Forsyth County, Georgia
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##	Foster County, North Dakota	Franklin city, Virginia
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##	Franklin County, Florida	Franklin County, Georgia
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##	Franklin County, Illinois	Franklin County, Iowa
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##	Franklin County, Kentucky	Franklin County, Missouri
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##	Franklin County, Nebraska	Franklin County, North Carolina
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##	Franklin County, Vermont	Franklin County, Virginia
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##	Franklin County, Washington	Frederick County, Maryland
##	1	1
##	Frederick County, Virginia	Fredericksburg city, Virginia
##	1	1
##	Freeborn County, Minnesota	Freestone County, Texas
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##	Fremont County, Idaho	Fremont County, Wyoming
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##	Fresno County, California	Frontier County, Nebraska
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##	Fulton County, Arkansas	Fulton County, New York
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##	Fulton County, Pennsylvania	Furnas County, Nebraska
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##	Gage County, Nebraska	Gallatin County, Illinois
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##	Gallatin County, Montana	Gallia County, Ohio
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##	Galveston County, Texas	Garden County, Nebraska
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##	Garfield County, Colorado	Garfield County, Oklahoma
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##	Garrard County, Kentucky	Garrett County, Maryland
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##	Gasconade County, Missouri	Gem County, Idaho
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##	Genesee County, Michigan	Geneva County, Alabama
##	1	1
##	George County, Mississippi	Gibson County, Indiana
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##	Giles County, Tennessee	Gillespie County, Texas
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##	Gilliam County, Oregon	1	Gilmer County, West Virginia	1
##	Gladwin County, Michigan	1	Glascocock County, Georgia	1
##	Glenn County, California	1	Gloucester County, Virginia	1
##	Glynn County, Georgia	1	Gogebic County, Michigan	1
##	Goliad County, Texas	1	Gonzales County, Texas	1
##	Goodhue County, Minnesota	1	Gordon County, Georgia	1
##	Goshen County, Wyoming	1	Gosper County, Nebraska	1
##	Grady County, Georgia	1	Grady County, Oklahoma	1
##	Graham County, North Carolina	1	Grand County, Colorado	1
##	Grand County, Utah	1	Grand Isle County, Vermont	1
##	Grant County, Arkansas	1	Grant County, Kansas	1
##	Grant County, Kentucky	1	Grant County, Minnesota	1
##	Grant County, Oklahoma	1	Grant County, Oregon	1
##	Grant County, West Virginia	1	Grant Parish, Louisiana	1
##	Granville County, North Carolina	1	Graves County, Kentucky	1
##	Gray County, Kansas	1	Grays Harbor County, Washington	1
##	Grayson County, Texas	1	Grayson County, Virginia	1
##	Greeley County, Kansas	1	Greeley County, Nebraska	1
##	Green County, Kentucky	1	Green County, Wisconsin	1
##	Green Lake County, Wisconsin	1	Greene County, Illinois	1
##	Greene County, Indiana	1	Greene County, Iowa	1
##	Greene County, Missouri	1	Greene County, North Carolina	1
##	Greene County, Pennsylvania	1	Greene County, Tennessee	1
##	Greene County, Virginia	1	Greenlee County, Arizona	1
##	Greenup County, Kentucky	1	Greenville County, South Carolina	1
##	Greenwood County, South Carolina	1	Gregg County, Texas	1
##	Grenada County, Mississippi	1	Griggs County, North Dakota	1

##	Grundy County, Illinois	1	Grundy County, Iowa	1
##	Grundy County, Missouri	1	Guadalupe County, New Mexico	1
##	Gulf County, Florida	1	Guthrie County, Iowa	1
##	Haakon County, South Dakota	1	Habersham County, Georgia	1
##	Haines Borough, Alaska	1	Hale County, Alabama	1
##	Hale County, Texas	1	Halifax County, North Carolina	1
##	Halifax County, Virginia	1	Hall County, Georgia	1
##	Hall County, Nebraska	1	Hall County, Texas	1
##	Hamblen County, Tennessee	1	Hamilton County, Illinois	1
##	Hamilton County, Indiana	1	Hamilton County, Iowa	1
##	Hamilton County, Kansas	1	Hamilton County, Nebraska	1
##	Hamilton County, New York	1	Hamilton County, Ohio	1
##	Hamilton County, Texas	1	Hamlin County, South Dakota	1
##	Hampshire County, West Virginia	1	Hampton County, South Carolina	1
##	Hancock County, Indiana	1	Hancock County, Iowa	1
##	Hancock County, Kentucky	1	Hancock County, Maine	1
##	Hancock County, Mississippi	1	Hancock County, Tennessee	1
##	Hancock County, West Virginia	1	Hansford County, Texas	1
##	Haralson County, Georgia	1	Hardin County, Iowa	1
##	Hardin County, Kentucky	1	Hardin County, Ohio	1
##	Hardy County, West Virginia	1	Harford County, Maryland	1
##	Harlan County, Kentucky	1	Harlan County, Nebraska	1
##	Harmon County, Oklahoma	1	Harper County, Oklahoma	1
##	Harris County, Texas	1	Harrison County, Iowa	1
##	Harrison County, Mississippi	1	Harrison County, Texas	1
##	Harrison County, West Virginia	1	Hart County, Kentucky	1
##	Hartford County, Connecticut	1	Hartley County, Texas	1

##	Harvey County, Kansas	Haskell County, Texas
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##	Hawkins County, Tennessee	Haywood County, North Carolina
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##	Heard County, Georgia	Hempstead County, Arkansas
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##	Henderson County, Illinois	Henderson County, North Carolina
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##	Henderson County, Tennessee	Hendricks County, Indiana
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##	Hennepin County, Minnesota	Henrico County, Virginia
##	1	1
##	Henry County, Georgia	Henry County, Illinois
##	1	1
##	Henry County, Indiana	Henry County, Iowa
##	1	1
##	Henry County, Missouri	Henry County, Tennessee
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##	Herkimer County, New York	Hertford County, North Carolina
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##	Hettinger County, North Dakota	Hickman County, Kentucky
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##	Hickman County, Tennessee	Hickory County, Missouri
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##	Hidalgo County, Texas	Highland County, Virginia
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##	Highlands County, Florida	Hill County, Texas
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##	Hillsborough County, New Hampshire	Hillsdale County, Michigan
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##	Hockley County, Texas	Hoke County, North Carolina
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##	Holmes County, Mississippi	Hood County, Texas
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##	Hood River County, Oregon	Hopewell city, Virginia
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##	Hopkins County, Texas	Horry County, South Carolina
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##	Hot Spring County, Arkansas	Houghton County, Michigan
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##	Houston County, Alabama	Houston County, Georgia
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##	Houston County, Minnesota	Houston County, Texas
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##	Howard County, Arkansas	Howard County, Iowa
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##	Howard County, Maryland	Howard County, Missouri
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##	Howard County, Nebraska	Howard County, Texas
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##	Howell County, Missouri	Huerfano County, Colorado
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##	Hughes County, Oklahoma	Hughes County, South Dakota
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##	Humboldt County, Nevada	Humphreys County, Mississippi
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##	Humphreys County, Tennessee	Hunt County, Texas
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##	Hunterdon County, New Jersey	Huntingdon County, Pennsylvania
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##	Huron County, Michigan	Hutchinson County, South Dakota
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##	Hutchinson County, Texas	Hyde County, North Carolina
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##	Iberia Parish, Louisiana	Ida County, Iowa
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##	Imperial County, California	Independence County, Arkansas
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##	Indian River County, Florida	Indiana County, Pennsylvania
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##	Ingham County, Michigan	Inyo County, California
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##	Iosco County, Michigan	Iowa County, Iowa
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##	Iowa County, Wisconsin	Iredell County, North Carolina
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##	Iron County, Missouri	Iron County, Utah
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##	Iroquois County, Illinois	Irwin County, Georgia
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##	Isanti County, Minnesota	Issaquena County, Mississippi
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##	Itasca County, Minnesota	Izard County, Arkansas
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##	Jack County, Texas	Jackson County, Arkansas
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##	Jackson County, Georgia	Jackson County, Indiana
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##	Jackson County, Iowa	Jackson County, Kansas
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##	Jackson County, Kentucky	Jackson County, Mississippi
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##	Jackson County, Missouri	Jackson County, North Carolina
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##	Jackson County, Oregon	Jackson County, Texas
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##	Jackson County, West Virginia	James City County, Virginia
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##	Jasper County, Illinois	Jasper County, Indiana
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##	Jasper County, Mississippi	Jasper County, Missouri
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##	Jay County, Indiana	Jeff Davis County, Texas
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##	Jefferson County, Arkansas	Jefferson County, Colorado
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##	Jefferson County, Georgia	Jefferson County, Idaho
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##	Jefferson County, Illinois	Jefferson County, Iowa
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##	Jefferson County, Kansas	Jefferson County, Mississippi
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##	Jefferson County, Montana	Jefferson County, Nebraska
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##	Jefferson County, New York	Jefferson County, Oklahoma
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##	Jefferson County, Pennsylvania	Jefferson County, Texas
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##	Jefferson County, Washington	Jefferson County, West Virginia
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##	Jefferson County, Wisconsin	Jefferson Davis County, Mississippi
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##	Jefferson Davis Parish, Louisiana	Jefferson Parish, Louisiana
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##	Jenkins County, Georgia	Jennings County, Indiana
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##	Jerauld County, South Dakota	Jerome County, Idaho
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##	Jersey County, Illinois	Jessamine County, Kentucky
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##	Jewell County, Kansas	Jim Hogg County, Texas
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##	Jo Daviess County, Illinois	Johnson County, Arkansas
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##	Johnson County, Georgia	Johnson County, Iowa
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##	Johnson County, Kansas	Johnson County, Nebraska
##	1	1
##	Johnson County, Tennessee	Johnston County, North Carolina
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##	Jones County, Georgia	Jones County, Iowa
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##	Jones County, Mississippi	Jones County, Texas
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##	Josephine County, Oregon	Juab County, Utah
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##	Judith Basin County, Montana	Juneau County, Wisconsin
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##	Kalamazoo County, Michigan	Kanabec County, Minnesota
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##	Kane County, Utah	Karnes County, Texas
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##	Kauai County, Hawaii	Kay County, Oklahoma
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##	Kearney County, Nebraska	Kearny County, Kansas
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##	Keith County, Nebraska	Kemper County, Mississippi
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##	Kendall County, Illinois	Kennebec County, Maine
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##	Kent County, Delaware	Kent County, Maryland
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##	Kent County, Rhode Island	1	Keokuk County, Iowa	1
##	Kern County, California	1	Kerr County, Texas	1
##	Kershaw County, South Carolina	1	Keweenaw County, Michigan	1
##	Kimball County, Nebraska	1	Kimble County, Texas	1
##	King and Queen County, Virginia	1	King William County, Virginia	1
##	Kinney County, Texas	1	Kiowa County, Colorado	1
##	Kiowa County, Kansas	1	Kit Carson County, Colorado	1
##	Kittson County, Minnesota	1	Knott County, Kentucky	1
##	Knox County, Illinois	1	Knox County, Indiana	1
##	Knox County, Kentucky	1	Knox County, Maine	1
##	Knox County, Missouri	1	Knox County, Ohio	1
##	Kodiak Island Borough, Alaska	1	Koochiching County, Minnesota	1
##	Kosciusko County, Indiana	1	La Paz County, Arizona	1
##	La Salle County, Texas	1	Lac qui Parle County, Minnesota	1
##	Lackawanna County, Pennsylvania	1	Lafayette County, Arkansas	1
##	Lafayette County, Mississippi	1	Lafayette County, Missouri	1
##	Lafayette County, Wisconsin	1	Lafourche Parish, Louisiana	1
##	Lake County, Florida	1	Lake County, Illinois	1
##	Lake County, Indiana	1	Lake County, Ohio	1
##	Lake County, South Dakota	1	Lake County, Tennessee	1
##	Lamar County, Alabama	1	Lamar County, Georgia	1
##	Lamar County, Mississippi	1	Lamoille County, Vermont	1
##	LaMoure County, North Dakota	1	Lampasas County, Texas	1
##	Lancaster County, Nebraska	1	Lancaster County, Pennsylvania	1
##	Lander County, Nevada	1	Larue County, Kentucky	1
##	Lassen County, California	1	Latimer County, Oklahoma	1
##	Lauderdale County, Alabama	1	Lauderdale County, Mississippi	1

##	Laurel County, Kentucky	Lawrence County, Alabama
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##	Lawrence County, Arkansas	Lawrence County, Illinois
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##	Lawrence County, Indiana	Lawrence County, Kentucky
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##	Lawrence County, Ohio	Lawrence County, Pennsylvania
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##	Lawrence County, South Dakota	Lawrence County, Tennessee
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##	Le Sueur County, Minnesota	Leake County, Mississippi
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##	Lebanon County, Pennsylvania	Lee County, Arkansas
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##	Lee County, Georgia	Lee County, Iowa
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##	Lee County, South Carolina	Lee County, Texas
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##	Lee County, Virginia	Leflore County, Mississippi
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##	Lehigh County, Pennsylvania	Lemhi County, Idaho
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##	Lenawee County, Michigan	Leon County, Florida
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##	Leslie County, Kentucky	Letcher County, Kentucky
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##	Lewis County, Idaho	Lewis County, Kentucky
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##	Lewis County, New York	Lewis County, Tennessee
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##	Lexington County, South Carolina	Liberty County, Florida
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##	Liberty County, Montana	Licking County, Ohio
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##	Limestone County, Alabama	Lincoln County, Arkansas
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##	Lincoln County, Colorado	Lincoln County, Georgia
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##	Lincoln County, Idaho	Lincoln County, Kentucky
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##	Lincoln County, Maine	Lincoln County, Minnesota
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##	Lincoln County, Mississippi	Lincoln County, Montana
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##	Lincoln County, Nebraska	Lincoln County, Nevada
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##	Lincoln County, Oklahoma	Lincoln County, Oregon
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##	Lincoln County, South Dakota	Lincoln County, Tennessee
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##	Lincoln County, Washington	Lincoln County, Wisconsin
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##	Lincoln County, Wyoming	Linn County, Kansas
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##	Linn County, Oregon	Lipscomb County, Texas
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##	Litchfield County, Connecticut	Live Oak County, Texas
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##	Livingston County, Illinois	Livingston County, Kentucky
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##	Livingston County, Michigan	Livingston County, Missouri
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##	Livingston County, New York	Llano County, Texas
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##	Logan County, Colorado	Logan County, Illinois
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##	Lorain County, Ohio	Los Alamos County, New Mexico
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##	Los Angeles County, California	Loudoun County, Virginia
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##	Louisa County, Virginia	Lowndes County, Alabama
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##	Lowndes County, Georgia	Lowndes County, Mississippi
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##	Luna County, New Mexico	Lunenburg County, Virginia
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##	Lycoming County, Pennsylvania	Lyman County, South Dakota
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##	Lynn County, Texas	Lyon County, Kansas
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##	Lyon County, Minnesota	Lyon County, Nevada
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##	Macomb County, Michigan	Macon County, Georgia
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##	Macon County, Illinois	Macon County, Tennessee
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##	Macoupin County, Illinois	Madison County, Alabama
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##	Madison County, Florida	Madison County, Indiana
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##	Madison County, Kentucky	Madison County, Mississippi
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##	Madison County, Missouri	Madison County, New York
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##	Madison County, North Carolina	Madison County, Ohio
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##	Madison County, Tennessee	Mahaska County, Iowa
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##	Mahnomen County, Minnesota	Malheur County, Oregon
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##	Manassas city, Virginia	Manitowoc County, Wisconsin
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##	Marathon County, Wisconsin	Maricopa County, Arizona
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##	Maries County, Missouri	Marinette County, Wisconsin
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##	Marion County, Arkansas	Marion County, Georgia
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##	Marion County, Illinois	Marion County, Indiana
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##	Marion County, Iowa	Marion County, Kansas
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##	Marion County, Kentucky	Marion County, Mississippi
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##	Marion County, Missouri	Marion County, Texas
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##	Marion County, West Virginia	Mariposa County, California
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##	Marlboro County, South Carolina	Marquette County, Wisconsin
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##	Marshall County, Alabama	Marshall County, Indiana
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##	Marshall County, Iowa	Marshall County, Kentucky
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##	Marshall County, Minnesota	Marshall County, Mississippi
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##	Marshall County, South Dakota	Marshall County, Tennessee
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##	Marshall County, West Virginia	Martin County, Florida
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##	Martin County, Indiana	Martin County, Kentucky
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##	Martin County, Minnesota	Martin County, North Carolina
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##	Martin County, Texas	Mason County, Kentucky
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##	Mason County, Michigan	Mason County, Texas
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##	Mason County, Washington	Mason County, West Virginia
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##	Massac County, Illinois	Matanuska-Susitna Borough, Alaska
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##	Maui County, Hawaii	Mauzy County, Tennessee
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##	Maverick County, Texas	Mayes County, Oklahoma
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##	McClain County, Oklahoma	McCormick County, South Carolina
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##	McCracken County, Kentucky	McCurtain County, Oklahoma
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##	McDonald County, Missouri	McDowell County, North Carolina
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##	McHenry County, North Dakota	McIntosh County, Georgia
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##	McIntosh County, North Dakota	McIntosh County, Oklahoma
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##	McKean County, Pennsylvania	McKenzie County, North Dakota
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##	McLean County, Illinois	McLean County, Kentucky
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##	McLean County, North Dakota	McLennan County, Texas
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##	McMinn County, Tennessee	McPherson County, Kansas
##	1	1
##	Meade County, Kansas	Meade County, Kentucky
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##	Meade County, South Dakota	Meagher County, Montana
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##	Mecklenburg County, North Carolina	Mecosta County, Michigan
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##	Meeker County, Minnesota	Meigs County, Tennessee
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##	Mellette County, South Dakota	Menard County, Illinois
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##	Menard County, Texas	Mendocino County, California
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##	Menifee County, Kentucky	Menominee County, Michigan
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##	Merced County, California	Mercer County, Kentucky
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##	Mercer County, West Virginia	Meriwether County, Georgia
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##	Merrick County, Nebraska	Merrimack County, New Hampshire
##	1	1
##	Mesa County, Colorado	Metcalf County, Kentucky
##	1	1
##	Miami-Dade County, Florida	Miami County, Kansas
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##	Miami County, Ohio	Middlesex County, Connecticut
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##	Middlesex County, Massachusetts	Mifflin County, Pennsylvania
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##	Milam County, Texas	Millard County, Utah
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##	Mille Lacs County, Minnesota	Miller County, Arkansas
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##	Miller County, Georgia	Mills County, Iowa
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##	Miner County, South Dakota	Mineral County, Nevada
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##	Mineral County, West Virginia	Mingo County, West Virginia
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##	Minidoka County, Idaho	Minnehaha County, South Dakota
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##	Mississippi County, Arkansas	Mississippi County, Missouri
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##	Mitchell County, Kansas	Mitchell County, North Carolina
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##	Mitchell County, Texas	Moniteau County, Missouri
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##	Monmouth County, New Jersey	Monona County, Iowa
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##	Monroe County, Alabama	Monroe County, Arkansas
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##	Monroe County, Florida	Monroe County, Georgia
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##	Monroe County, Indiana	Monroe County, Kentucky
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##	Monroe County, Mississippi	Monroe County, New York
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##	Monroe County, Ohio	Monroe County, Pennsylvania
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##	Monroe County, Tennessee	Monroe County, West Virginia
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##	Monroe County, Wisconsin	Montague County, Texas
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##	Montcalm County, Michigan	Monterey County, California
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##	Montgomery County, Alabama	Montgomery County, Georgia
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##	Montgomery County, Illinois	Montgomery County, Kansas
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##	Montgomery County, Kentucky	Montgomery County, Maryland
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##	Montgomery County, Mississippi	Montgomery County, Missouri
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##	Montgomery County, New York	Montgomery County, North Carolina
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##	Montgomery County, Pennsylvania	Montgomery County, Tennessee
##	1	1
##	Montgomery County, Texas	Montgomery County, Virginia
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##	Montmorency County, Michigan	Montrose County, Colorado
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##	Moore County, Tennessee	Moore County, Texas
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##	Mora County, New Mexico	Morgan County, Alabama
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##	Morgan County, Colorado	Morgan County, Georgia
##	1	1
##	Morgan County, Illinois	Morgan County, Indiana
##	1	1
##	Morgan County, Missouri	Morgan County, West Virginia
##	1	1
##	Morrill County, Nebraska	Morris County, New Jersey
##	1	1
##	Morrow County, Ohio	Morton County, Kansas
##	1	1
##	Moultrie County, Illinois	Mountrail County, North Dakota
##	1	1
##	Multnomah County, Oregon	Muscogee County, Georgia
##	1	1
##	Muskogee County, Oklahoma	Nacogdoches County, Texas
##	1	1
##	Nash County, North Carolina	Nassau County, Florida
##	1	1
##	Nassau County, New York	Natchitoches Parish, Louisiana
##	1	1
##	Natrona County, Wyoming	Nelson County, Virginia
##	1	1

##	Nemaha County, Nebraska	Neosho County, Kansas
##	1	1
##	Neshoba County, Mississippi	Ness County, Kansas
##	1	1
##	Nevada County, Arkansas	Nevada County, California
##	1	1
##	New Haven County, Connecticut	New Kent County, Virginia
##	1	1
##	New London County, Connecticut	New Madrid County, Missouri
##	1	1
##	Newaygo County, Michigan	Newport County, Rhode Island
##	1	1
##	Newton County, Georgia	Newton County, Missouri
##	1	1
##	Newton County, Texas	Nez Perce County, Idaho
##	1	1
##	Niagara County, New York	Nicollet County, Minnesota
##	1	1
##	Niobrara County, Wyoming	Noble County, Indiana
##	1	1
##	Noble County, Ohio	Noble County, Oklahoma
##	1	1
##	Nobles County, Minnesota	Nodaway County, Missouri
##	1	1
##	Nolan County, Texas	Nome Census Area, Alaska
##	1	1
##	Norfolk County, Massachusetts	Northumberland County, Pennsylvania
##	1	1
##	Northwest Arctic Borough, Alaska	Norton city, Virginia
##	1	1
##	Nottoway County, Virginia	Noxubee County, Mississippi
##	1	1
##	Nuckolls County, Nebraska	Nueces County, Texas
##	1	1
##	Oakland County, Michigan	Obion County, Tennessee
##	1	1
##	Ocean County, New Jersey	Oceana County, Michigan
##	1	1
##	Oconee County, South Carolina	Oconto County, Wisconsin
##	1	1
##	Ogemaw County, Michigan	Oglethorpe County, Georgia
##	1	1
##	Ohio County, Indiana	Ohio County, Kentucky
##	1	1
##	Ohio County, West Virginia	Okaloosa County, Florida
##	1	1
##	Okeechobee County, Florida	Okfuskee County, Oklahoma
##	1	1
##	Oktibbeha County, Mississippi	Oldham County, Kentucky
##	1	1
##	Oneida County, Idaho	Oneida County, New York
##	1	1
##	Onondaga County, New York	Onslow County, North Carolina
##	1	1

##	Ontario County, New York	1	Ontonagon County, Michigan	1
##	Orange County, California	1	Orange County, Florida	1
##	Orange County, New York	1	Orange County, North Carolina	1
##	Orange County, Vermont	1	Orange County, Virginia	1
##	Osage County, Oklahoma	1	Osborne County, Kansas	1
##	Osceola County, Florida	1	Oscoda County, Michigan	1
##	Otero County, Colorado	1	Otero County, New Mexico	1
##	Otoe County, Nebraska	1	Otsego County, Michigan	1
##	Ottawa County, Kansas	1	Ottawa County, Michigan	1
##	Ottawa County, Ohio	1	Ottawa County, Oklahoma	1
##	Otter Tail County, Minnesota	1	Ouachita County, Arkansas	1
##	Ouray County, Colorado	1	Overton County, Tennessee	1
##	Owen County, Indiana	1	Ozaukee County, Wisconsin	1
##	Pacific County, Washington	1	Page County, Iowa	1
##	Page County, Virginia	1	Palm Beach County, Florida	1
##	Pamlico County, North Carolina	1	Panola County, Mississippi	1
##	Park County, Colorado	1	Park County, Montana	1
##	Park County, Wyoming	1	Parke County, Indiana	1
##	Parker County, Texas	1	Pasco County, Florida	1
##	Pasquotank County, North Carolina	1	Patrick County, Virginia	1
##	Pawnee County, Nebraska	1	Pawnee County, Oklahoma	1
##	Payette County, Idaho	1	Payne County, Oklahoma	1
##	Peach County, Georgia	1	Pearl River County, Mississippi	1
##	Pecos County, Texas	1	Pembina County, North Dakota	1
##	Pender County, North Carolina	1	Pendleton County, Kentucky	1
##	Pennington County, Minnesota	1	Penobscot County, Maine	1
##	Peoria County, Illinois	1	Pepin County, Wisconsin	1

##	Perkins County, Nebraska	Perkins County, South Dakota
##	1	1
##	Perquimans County, North Carolina	Perry County, Illinois
##	1	1
##	Perry County, Indiana	Perry County, Kentucky
##	1	1
##	Perry County, Mississippi	Perry County, Missouri
##	1	1
##	Perry County, Ohio	Person County, North Carolina
##	1	1
##	Pettis County, Missouri	Phelps County, Missouri
##	1	1
##	Philadelphia County, Pennsylvania	Phillips County, Colorado
##	1	1
##	Pickaway County, Ohio	Pickens County, Alabama
##	1	1
##	Pickens County, Georgia	Pickett County, Tennessee
##	1	1
##	Pierce County, Georgia	Pierce County, Washington
##	1	1
##	Pike County, Georgia	Pike County, Illinois
##	1	1
##	Pike County, Indiana	Pike County, Mississippi
##	1	1
##	Pike County, Ohio	Pike County, Pennsylvania
##	1	1
##	Pima County, Arizona	Pine County, Minnesota
##	1	1
##	Pinellas County, Florida	Pitkin County, Colorado
##	1	1
##	Pitt County, North Carolina	Pittsylvania County, Virginia
##	1	1
##	Plaquemines Parish, Louisiana	Platte County, Missouri
##	1	1
##	Platte County, Nebraska	Platte County, Wyoming
##	1	1
##	Pleasants County, West Virginia	Plymouth County, Massachusetts
##	1	1
##	Pocahontas County, Iowa	Poinsett County, Arkansas
##	1	1
##	Pointe Coupee Parish, Louisiana	Polk County, Florida
##	1	1
##	Polk County, Iowa	Polk County, Missouri
##	1	1
##	Polk County, Nebraska	Polk County, North Carolina
##	1	1
##	Polk County, Texas	Pondera County, Montana
##	1	1
##	Pope County, Illinois	Portage County, Ohio
##	1	1
##	Portage County, Wisconsin	Posey County, Indiana
##	1	1
##	Pottawatomie County, Kansas	Pottawatomie County, Oklahoma
##	1	1

##	Pottawattamie County, Iowa	Potter County, Pennsylvania
##	1	1
##	Powell County, Kentucky	Power County, Idaho
##	1	1
##	Poweshiek County, Iowa	Powhatan County, Virginia
##	1	1
##	Prairie County, Arkansas	Pratt County, Kansas
##	1	1
##	Preble County, Ohio	Prentiss County, Mississippi
##	1	1
##	Presidio County, Texas	Presque Isle County, Michigan
##	1	1
##	Preston County, West Virginia	Prince George County, Virginia
##	1	1
##	Prince William County, Virginia	Prowers County, Colorado
##	1	1
##	Pueblo County, Colorado	Pulaski County, Arkansas
##	1	1
##	Pulaski County, Georgia	Pulaski County, Indiana
##	1	1
##	Pulaski County, Kentucky	Pulaski County, Missouri
##	1	1
##	Putnam County, Georgia	Putnam County, Illinois
##	1	1
##	Putnam County, Indiana	Putnam County, Tennessee
##	1	1
##	Quay County, New Mexico	Queen Anne's County, Maryland
##	1	1
##	Queens County, New York	Quitman County, Georgia
##	1	1
##	Quitman County, Mississippi	Racine County, Wisconsin
##	1	1
##	Rains County, Texas	Randall County, Texas
##	1	1
##	Randolph County, Alabama	Randolph County, Arkansas
##	1	1
##	Randolph County, Georgia	Randolph County, North Carolina
##	1	1
##	Randolph County, West Virginia	Rankin County, Mississippi
##	1	1
##	Ransom County, North Dakota	Rapides Parish, Louisiana
##	1	1
##	Rappahannock County, Virginia	Ravalli County, Montana
##	1	1
##	Rawlins County, Kansas	Ray County, Missouri
##	1	1
##	Real County, Texas	Red Willow County, Nebraska
##	1	1
##	Redwood County, Minnesota	Reeves County, Texas
##	1	1
##	Reno County, Kansas	Rensselaer County, New York
##	1	1
##	Renville County, North Dakota	Republic County, Kansas
##	1	1

##	Rhea County, Tennessee	1	Rice County, Kansas	1
##	Richardson County, Nebraska	1	Richland County, Illinois	1
##	Richland County, Ohio	1	Richland County, South Carolina	1
##	Richland Parish, Louisiana	1	Richmond County, New York	1
##	Richmond County, North Carolina	1	Rio Arriba County, New Mexico	1
##	Ripley County, Indiana	1	Roane County, Tennessee	1
##	Roane County, West Virginia	1	Roanoke County, Virginia	1
##	Roberts County, South Dakota	1	Robeson County, North Carolina	1
##	Rock County, Minnesota	1	Rock County, Wisconsin	1
##	Rock Island County, Illinois	1	Rockbridge County, Virginia	1
##	Rockingham County, North Carolina	1	Rockingham County, Virginia	1
##	Rockland County, New York	1	Rogers County, Oklahoma	1
##	Rooks County, Kansas	1	Roosevelt County, Montana	1
##	Roscommon County, Michigan	1	Roseau County, Minnesota	1
##	Ross County, Ohio	1	Rowan County, Kentucky	1
##	Runnels County, Texas	1	Rush County, Kansas	1
##	Rusk County, Texas	1	Rusk County, Wisconsin	1
##	Russell County, Alabama	1	Russell County, Kentucky	1
##	Russell County, Virginia	1	Rutherford County, North Carolina	1
##	Rutherford County, Tennessee	1	Sabine County, Texas	1
##	Sabine Parish, Louisiana	1	Sacramento County, California	1
##	Sagadahoc County, Maine	1	Saginaw County, Michigan	1
##	Salem city, Virginia	1	Salem County, New Jersey	1
##	Saline County, Illinois	1	Saline County, Nebraska	1
##	Salt Lake County, Utah	1	Saluda County, South Carolina	1
##	Sampson County, North Carolina	1	San Augustine County, Texas	1
##	San Benito County, California	1	San Jacinto County, Texas	1

##	San Juan County, Utah	San Juan County, Washington
##	1	1
##	San Luis Obispo County, California	San Mateo County, California
##	1	1
##	San Miguel County, Colorado	San Miguel County, New Mexico
##	1	1
##	San Patricio County, Texas	San Saba County, Texas
##	1	1
##	Sanborn County, South Dakota	Sanders County, Montana
##	1	1
##	Sandusky County, Ohio	Sanpete County, Utah
##	1	1
##	Santa Barbara County, California	Santa Clara County, California
##	1	1
##	Santa Cruz County, Arizona	Santa Cruz County, California
##	1	1
##	Santa Fe County, New Mexico	Santa Rosa County, Florida
##	1	1
##	Sauk County, Wisconsin	Saunders County, Nebraska
##	1	1
##	Sawyer County, Wisconsin	Schenectady County, New York
##	1	1
##	Schleicher County, Texas	Schley County, Georgia
##	1	1
##	Schoharie County, New York	Schuyler County, New York
##	1	1
##	Schuylkill County, Pennsylvania	Scioto County, Ohio
##	1	1
##	Scotland County, Missouri	Scott County, Arkansas
##	1	1
##	Scott County, Illinois	Scott County, Iowa
##	1	1
##	Scott County, Kansas	Scott County, Virginia
##	1	1
##	Scurry County, Texas	Searcy County, Arkansas
##	1	1
##	Sebastian County, Arkansas	Sedgwick County, Colorado
##	1	1
##	Seminole County, Georgia	Seminole County, Oklahoma
##	1	1
##	Seneca County, New York	Seneca County, Ohio
##	1	1
##	Sequatchie County, Tennessee	Sequoyah County, Oklahoma
##	1	1
##	Seward County, Kansas	Sharkey County, Mississippi
##	1	1
##	Shawano County, Wisconsin	Shawnee County, Kansas
##	1	1
##	Shelby County, Alabama	Shelby County, Illinois
##	1	1
##	Shelby County, Kentucky	Shelby County, Missouri
##	1	1
##	Shelby County, Tennessee	Shenandoah County, Virginia
##	1	1

##	Sheridan County, Kansas	Sheridan County, Montana
##	1	1
##	Sheridan County, North Dakota	Sherman County, Nebraska
##	1	1
##	Sherman County, Oregon	Shiawassee County, Michigan
##	1	1
##	Sierra County, New Mexico	Simpson County, Mississippi
##	1	1
##	Siskiyou County, California	Sitka City and Borough, Alaska
##	1	1
##	Skagit County, Washington	Skamania County, Washington
##	1	1
##	Smith County, Kansas	Smith County, Mississippi
##	1	1
##	Smith County, Tennessee	Smith County, Texas
##	1	1
##	Smyth County, Virginia	Snohomish County, Washington
##	1	1
##	Snyder County, Pennsylvania	Solano County, California
##	1	1
##	Somerset County, Maine	Somerset County, New Jersey
##	1	1
##	Somerset County, Pennsylvania	Sonoma County, California
##	1	1
##	Southampton County, Virginia	Spalding County, Georgia
##	1	1
##	Spartanburg County, South Carolina	Spencer County, Indiana
##	1	1
##	Spencer County, Kentucky	Spink County, South Dakota
##	1	1
##	St. Charles Parish, Louisiana	St. Clair County, Alabama
##	1	1
##	St. Clair County, Michigan	St. Francois County, Missouri
##	1	1
##	St. Helena Parish, Louisiana	St. James Parish, Louisiana
##	1	1
##	St. John the Baptist Parish, Louisiana	St. Johns County, Florida
##	1	1
##	St. Joseph County, Indiana	St. Landry Parish, Louisiana
##	1	1
##	St. Lawrence County, New York	St. Lucie County, Florida
##	1	1
##	St. Martin Parish, Louisiana	St. Mary's County, Maryland
##	1	1
##	St. Mary Parish, Louisiana	St. Tammany Parish, Louisiana
##	1	1
##	Stafford County, Kansas	Stafford County, Virginia
##	1	1
##	Stanley County, South Dakota	Stanly County, North Carolina
##	1	1
##	Stanton County, Nebraska	Starke County, Indiana
##	1	1
##	Starr County, Texas	Staunton city, Virginia
##	1	1

##	Ste. Genevieve County, Missouri	1	Stearns County, Minnesota	1
##	Steele County, Minnesota	1	Steele County, North Dakota	1
##	Stephens County, Georgia	1	Stephens County, Oklahoma	1
##	Stephens County, Texas	1	Steuben County, Indiana	1
##	Stevens County, Kansas	1	Stevens County, Washington	1
##	Stewart County, Georgia	1	Stillwater County, Montana	1
##	Stoddard County, Missouri	1	Stone County, Missouri	1
##	Stonewall County, Texas	1	Storey County, Nevada	1
##	Story County, Iowa	1	Strafford County, New Hampshire	1
##	Stutsman County, North Dakota	1	Suffolk city, Virginia	1
##	Suffolk County, Massachusetts	1	Sullivan County, Missouri	1
##	Sullivan County, New York	1	Sullivan County, Pennsylvania	1
##	Sullivan County, Tennessee	1	Sully County, South Dakota	1
##	Summers County, West Virginia	1	Summit County, Utah	1
##	Sumner County, Kansas	1	Sumner County, Tennessee	1
##	Sumter County, Georgia	1	Sunflower County, Mississippi	1
##	Surry County, Virginia	1	Susquehanna County, Pennsylvania	1
##	Suwannee County, Florida	1	Sweetwater County, Wyoming	1
##	Swift County, Minnesota	1	Swisher County, Texas	1
##	Switzerland County, Indiana	1	Talbot County, Maryland	1
##	Taliaferro County, Georgia	1	Tallahatchie County, Mississippi	1
##	Tama County, Iowa	1	Taos County, New Mexico	1
##	Tate County, Mississippi	1	Taylor County, Florida	1
##	Taylor County, Iowa	1	Taylor County, Wisconsin	1
##	Terrebonne Parish, Louisiana	1	Terry County, Texas	1
##	Teton County, Idaho	1	Texas County, Oklahoma	1
##	Thayer County, Nebraska	1	Thomas County, Georgia	1

##	Throckmorton County, Texas	Thurston County, Nebraska
##	1	1
##	Tift County, Georgia	Tioga County, Pennsylvania
##	1	1
##	Tippah County, Mississippi	Tippecanoe County, Indiana
##	1	1
##	Tipton County, Indiana	Tishomingo County, Mississippi
##	1	1
##	Todd County, Minnesota	Tolland County, Connecticut
##	1	1
##	Tompkins County, New York	Tooele County, Utah
##	1	1
##	Toombs County, Georgia	Torrance County, New Mexico
##	1	1
##	Towns County, Georgia	Traill County, North Dakota
##	1	1
##	Traverse County, Minnesota	Trego County, Kansas
##	1	1
##	Trigg County, Kentucky	Tripp County, South Dakota
##	1	1
##	Trousdale County, Tennessee	Trumbull County, Ohio
##	1	1
##	Tucker County, West Virginia	Tulsa County, Oklahoma
##	1	1
##	Tuolumne County, California	Turner County, Georgia
##	1	1
##	Turner County, South Dakota	Tuscaloosa County, Alabama
##	1	1
##	Tuscarawas County, Ohio	Twiggs County, Georgia
##	1	1
##	Twin Falls County, Idaho	Tyler County, Texas
##	1	1
##	Tyrrell County, North Carolina	Umatilla County, Oregon
##	1	1
##	Unicoi County, Tennessee	Union County, Florida
##	1	1
##	Union County, Georgia	Union County, Iowa
##	1	1
##	Union County, Kentucky	Union County, New Jersey
##	1	1
##	Union County, New Mexico	Union County, North Carolina
##	1	1
##	Union County, Ohio	Union County, Oregon
##	1	1
##	Union County, Pennsylvania	Union County, South Carolina
##	1	1
##	Union County, South Dakota	Upson County, Georgia
##	1	1
##	Upton County, Texas	Utah County, Utah
##	1	1
##	Uvalde County, Texas	Val Verde County, Texas
##	1	1
##	Valencia County, New Mexico	Valley County, Idaho
##	1	1

##	Van Buren County, Michigan	Van Buren County, Tennessee
##	1	1
##	Van Wert County, Ohio	Van Zandt County, Texas
##	1	1
##	Vermilion County, Illinois	Vermilion Parish, Louisiana
##	1	1
##	Vermillion County, Indiana	Vernon County, Missouri
##	1	1
##	Vernon County, Wisconsin	Vernon Parish, Louisiana
##	1	1
##	Vigo County, Indiana	Vilas County, Wisconsin
##	1	1
##	Virginia Beach city, Virginia	Wabash County, Illinois
##	1	1
##	Wabaunsee County, Kansas	Wadena County, Minnesota
##	1	1
##	Wagoner County, Oklahoma	Wake County, North Carolina
##	1	1
##	Wakulla County, Florida	Walla Walla County, Washington
##	1	1
##	Walsh County, North Dakota	Walthall County, Mississippi
##	1	1
##	Walton County, Georgia	Walworth County, South Dakota
##	1	1
##	Wapello County, Iowa	Ward County, North Dakota
##	1	1
##	Ware County, Georgia	Warren County, Illinois
##	1	1
##	Warren County, Indiana	Warren County, Iowa
##	1	1
##	Warren County, Kentucky	Warren County, Mississippi
##	1	1
##	Warren County, New Jersey	Warren County, North Carolina
##	1	1
##	Warren County, Ohio	Warren County, Pennsylvania
##	1	1
##	Warrick County, Indiana	Wasatch County, Utah
##	1	1
##	Washakie County, Wyoming	Washburn County, Wisconsin
##	1	1
##	Washington County, Alabama	Washington County, Arkansas
##	1	1
##	Washington County, Colorado	Washington County, Idaho
##	1	1
##	Washington County, Illinois	Washington County, Indiana
##	1	1
##	Washington County, Kansas	Washington County, Kentucky
##	1	1
##	Washington County, Maryland	Washington County, Minnesota
##	1	1
##	Washington County, Missouri	Washington County, Nebraska
##	1	1
##	Washington County, North Carolina	Washington County, Ohio
##	1	1

##	Washington County, Texas	Washington County, Virginia
##	1	1
##	Washington County, Wisconsin	Washington Parish, Louisiana
##	1	1
##	Washita County, Oklahoma	Waushara County, Wisconsin
##	1	1
##	Wayne County, Georgia	Wayne County, Illinois
##	1	1
##	Wayne County, Iowa	Wayne County, Michigan
##	1	1
##	Wayne County, Mississippi	Wayne County, Missouri
##	1	1
##	Wayne County, New York	Wayne County, Ohio
##	1	1
##	Wayne County, Pennsylvania	Wayne County, Tennessee
##	1	1
##	Wayne County, West Virginia	Waynesboro city, Virginia
##	1	1
##	Weakley County, Tennessee	Webb County, Texas
##	1	1
##	Webster County, Missouri	Webster County, Nebraska
##	1	1
##	Webster County, West Virginia	Webster Parish, Louisiana
##	1	1
##	Weld County, Colorado	Wells County, Indiana
##	1	1
##	Wells County, North Dakota	West Carroll Parish, Louisiana
##	1	1
##	Westchester County, New York	Wetzel County, West Virginia
##	1	1
##	Wharton County, Texas	Whatcom County, Washington
##	1	1
##	Wheatland County, Montana	Wheeler County, Oregon
##	1	1
##	White County, Illinois	White Pine County, Nevada
##	1	1
##	Whiteside County, Illinois	Whitfield County, Georgia
##	1	1
##	Whitley County, Indiana	Whitley County, Kentucky
##	1	1
##	Whitman County, Washington	Wichita County, Kansas
##	1	1
##	Wilbarger County, Texas	Wilkes County, Georgia
##	1	1
##	Wilkes County, North Carolina	Wilkin County, Minnesota
##	1	1
##	Wilkinson County, Georgia	Wilkinson County, Mississippi
##	1	1
##	Will County, Illinois	Willacy County, Texas
##	1	1
##	Williams County, North Dakota	Williams County, Ohio
##	1	1
##	Williamsburg city, Virginia	Williamsburg County, South Carolina
##	1	1

##	Williamson County, Illinois	Williamson County, Tennessee
##	1	1
##	Wilson County, Kansas	Wilson County, Texas
##	1	1
##	Windham County, Vermont	Winkler County, Texas
##	1	1
##	Winn Parish, Louisiana	Winnebago County, Illinois
##	1	1
##	Wise County, Texas	Wise County, Virginia
##	1	1
##	Wolfe County, Kentucky	Wood County, Ohio
##	1	1
##	Wood County, West Virginia	Wood County, Wisconsin
##	1	1
##	Woodford County, Kentucky	Woodruff County, Arkansas
##	1	1
##	Woods County, Oklahoma	Woodward County, Oklahoma
##	1	1
##	Worcester County, Massachusetts	Worth County, Missouri
##	1	1
##	Wright County, Minnesota	Wyandot County, Ohio
##	1	1
##	Wyandotte County, Kansas	Wyoming County, Pennsylvania
##	1	1
##	Yamhill County, Oregon	Yancey County, North Carolina
##	1	1
##	Yankton County, South Dakota	Yates County, New York
##	1	1
##	Yell County, Arkansas	Yellowstone County, Montana
##	1	1
##	Yoakum County, Texas	Yolo County, California
##	1	1
##	York County, Nebraska	York County, Pennsylvania
##	1	1
##	York County, South Carolina	York County, Virginia
##	1	1
##	Yuba County, California	Yukon-Koyukuk Census Area, Alaska
##	1	1
##	Yuma County, Arizona	Yuma County, Colorado
##	1	1
##	Zavala County, Texas	
##	1	

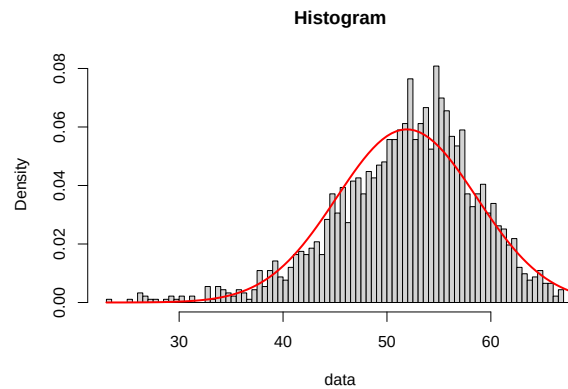
Variable *percentmarried*

Numerical variable that represent the percent of county residents who are married.

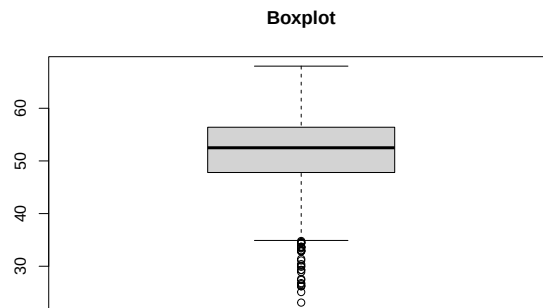
It has a non-symmetric distribution, it is skewed with a long left tail observed in the histogram. It doesn't have missing values, it has 34 outliers (non of them severe) and they are concentrated on the lower end of the distribution. With the shapiro-test we can confirm the rejection of the null hypothesis, therefore it doesn't have a normal distribution.

```
variable <- train$percentmarried
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      23.1   47.8   52.5   51.9   56.4   68.0
## [1] "Mean"
## [1] 51.90087
## [1] "Standard deviation"
## [1] 6.740129
```



```
## [1] "Skewness"
## [1] -0.6474867
```



```
## [1] "Num Outliers:"
## [1] 34
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.97753, p-value = 2.346e-16
```

```
outliers_df[res.var,"percentmarried"] <- 1
```

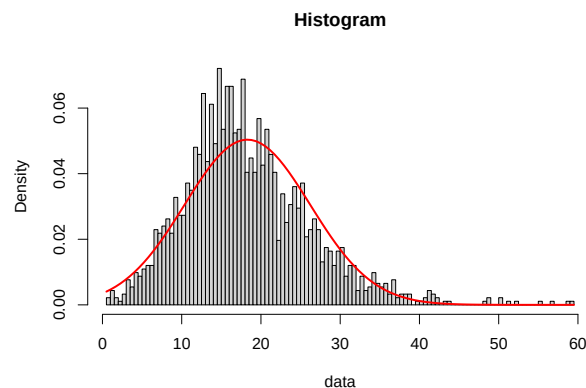
Variable *pctnohs18_24*

Numerical variable that show the percentage of county residents aged 18-24 whose highest level of education is less than a high school diploma.

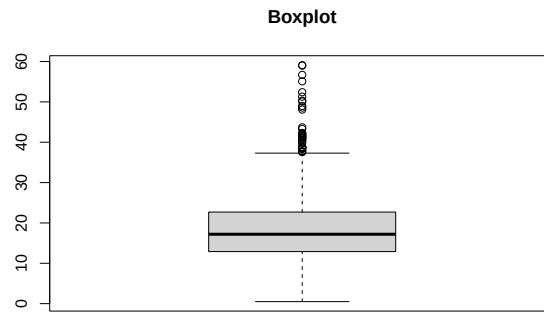
The distribution of this variable is skewed with a long right tail, we can confirm that it doesn't have a normal distribution thanks to the histogram and the curve. Moreover the p-value obtained in the shapiro-test confirm the non normal distribution. It doesn't have missing values, and it has 35 outliers, 5 of them severe.

```
variable <- train$pctnohs18_24  
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      0.50  12.90   17.20   18.29  22.70   59.10  
## [1] "Mean"  
## [1] 18.28777  
## [1] "Standard deviation"  
## [1] 7.92004
```



```
## [1] "Skewness"  
## [1] 0.8925103
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 5
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.96205, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctnohs18_24"] <- 1
```

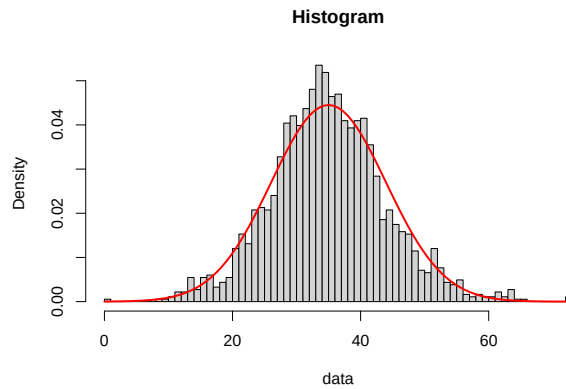
Variable *ppcths18_24*

Numeric variable, it represent the percentage of county residents ages 18-24 highest education attained is a high school diploma.

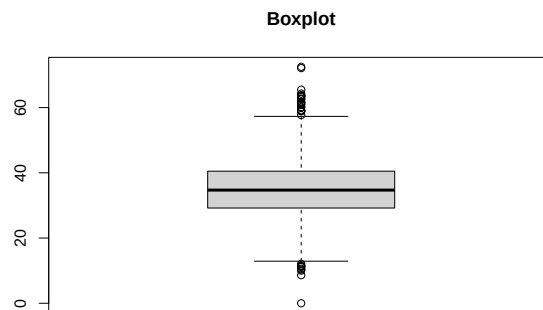
We can observe with the median(34,7%) and the mean(35%) that the variable has a symmetric distribution, but after the shapiro_test we observe that it doesn't have a normal distribution. It doesn't have missing values and it has 33 outliers, non of them severe.

```
variable <- train$ppcths18_24
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0   29.2    34.7    35.0   40.5    72.5
## [1] "Mean"
## [1] 34.99978
## [1] "Standard deviation"
## [1] 8.964462
```



```
## [1] "Skewness"
## [1] 0.2205398
```



```
## [1] "Num Outliers:"
## [1] 33
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99323, p-value = 1.922e-07
```

```
outliers_df[res.var,"pcths18_24"] <- 1
```

Variable *pctsomecol18_24*

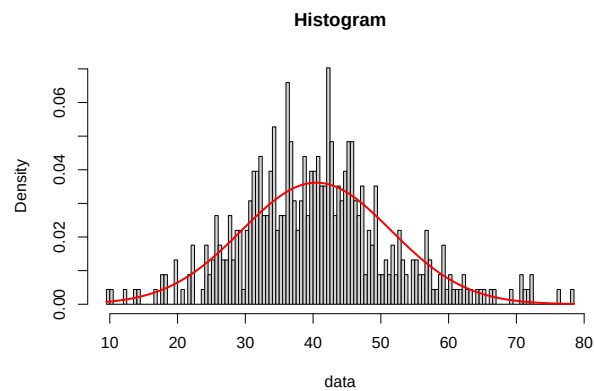
Numerical variable. Percent of county residents ages 18-24 highest education attained some college.

It has 1375 missing values. (75%) therefore we are going to do further actions in the imputation part. We will obtain the values knowing that $pctnohs18_24 + pcths18_24 + pctsomecol18_24 + pctbachdeg18_24 = 100$

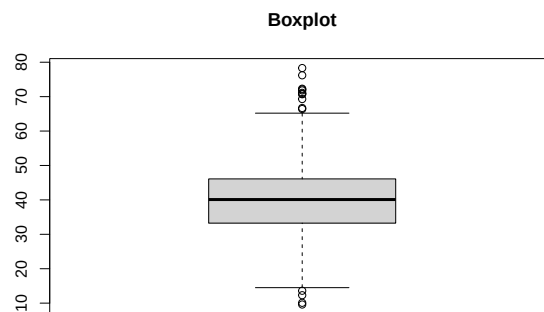
– A: Hauriem de fer els calculs aquí i ja calcular els outliers i tal be o ho fem a la part de imputation com esta ara?

```
variable <- train$pctsomecol18_24  
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's  
##      9.60  33.25   40.10   40.48  46.10   78.30   1376  
## [1] "Mean"  
## [1] 40.47846  
## [1] "Standard deviation"  
## [1] 11.0307
```



```
## [1] "Skewness"  
## [1] 0.4010929
```



```
## [1] "Num Outliers:"  
## [1] 15  
## [1] "Num. severe outliers:"  
## [1] 0  
## [1] "Shapiro test: "  
##
```

```
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98569, p-value = 0.0001868
```

```
outliers_df[res.var,"pctsomecol18_24"] <- 1
outliers <- which(outliers_df$pctsomecol18_24 == 1); length(outliers)
```

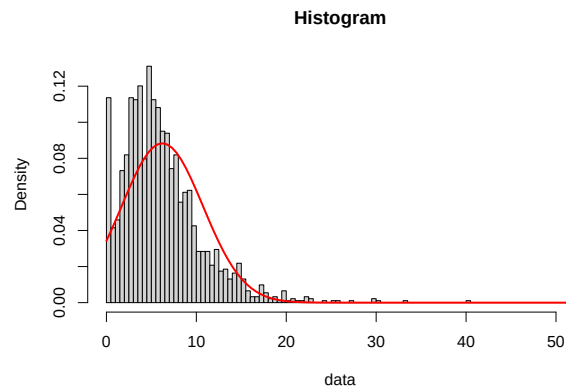
```
## [1] 15
```

Variable *pctbachdeg18_24*

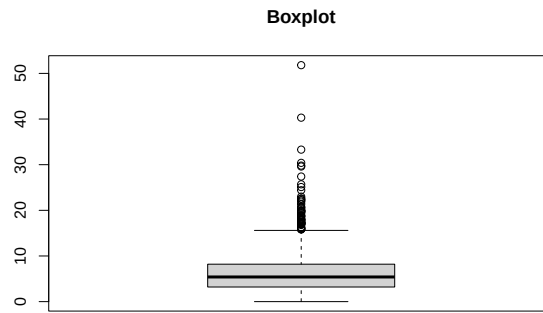
Numerical variable. Percent of county residents ages 18-24 highest education attained is a bachelor's degree. There are no missing values, it has 56 mild outliers 10 of them severes. After the shapiro-test we can reject the null hypothesis, thus it doesn't have a normal distribution.

```
variable <- train$pctbachdeg18_24
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.000  3.200   5.400   6.216  8.200  51.800
## [1] "Mean"
## [1] 6.216221
## [1] "Standard deviation"
## [1] 4.517556
```



```
## [1] "Skewness"
## [1] 1.980584
```



```
## [1] "Num Outliers:"
## [1] 56
## [1] "Num. severe outliers:"
## [1] 10
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.87959, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctbachdeg18_24"] <- 1
```

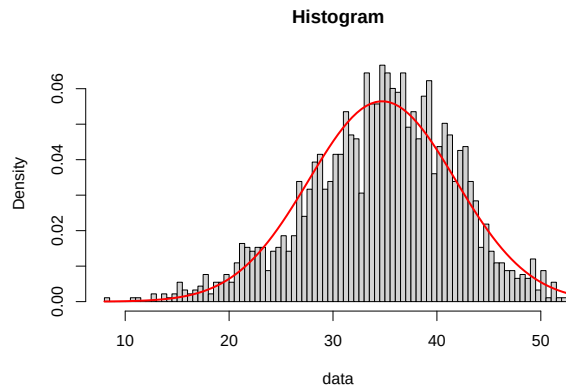
Variable *pcths25_over*

Numerical variable that express the percentage of county residents ages 25 and over highest education attained a high school diploma.

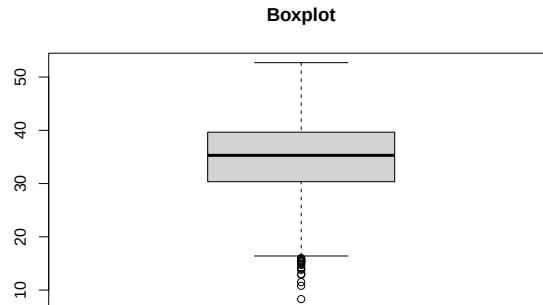
It doesn't have missing values but it have 18 outliers, non of the severe. It doesn't have a normal distribution.

```
variable <- train$pcths25_over
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      8.30  30.35   35.30   34.73   39.65   52.70
## [1] "Mean"
## [1] 34.73255
## [1] "Standard deviation"
## [1] 7.06599
```



```
## [1] "Skewness"
## [1] -0.3660496
```



```
## [1] "Num Outliers:"
## [1] 18
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99107, p-value = 3.741e-09
```

```
outliers_df[res.var, "pcths25_over"] <- 1
```

Variable *pctbachdeg25_over*

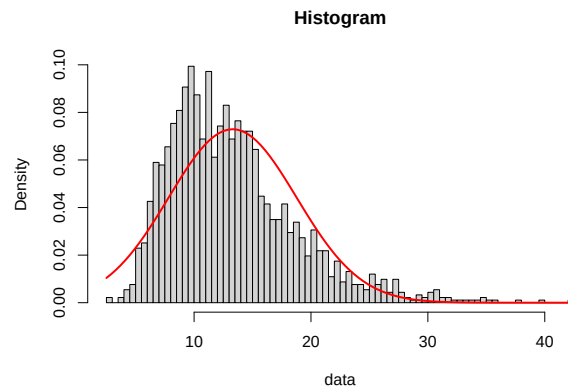
Numerical variable that expresses the the percentage of country resident ages 25 and over highest education attained: Bachelor's degree.

Firstly we can see that there is a huge left skewness and there is a relatively large number of outliers (62), from which just 3 are severe. From the plot we can already tell that no normal distribution is followed and a shapiro test over the variable confirms it.

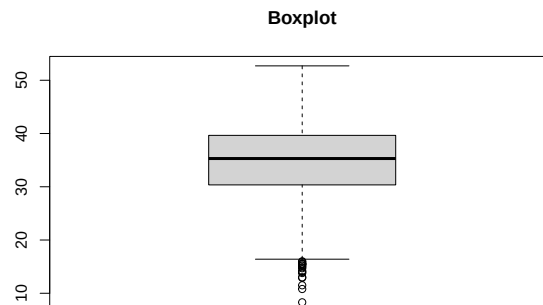
– op2 : mild = 59 | sev = 3

```
res.var <- summary_and_normality_check(train$pctbachdeg25_over)
```

```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##    2.5     9.3    12.3    13.3    16.0    42.2  
## [1] "Mean"  
## [1] 13.29754  
## [1] "Standard deviation"  
## [1] 5.470341
```



```
## [1] "Skewness"  
## [1] 1.159947
```



```
## [1] "Num Outliers:"  
## [1] 59  
## [1] "Num. severe outliers:"  
## [1] 3  
## [1] "Shapiro test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.92998, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctbachdeg25_over"] <- 1
```

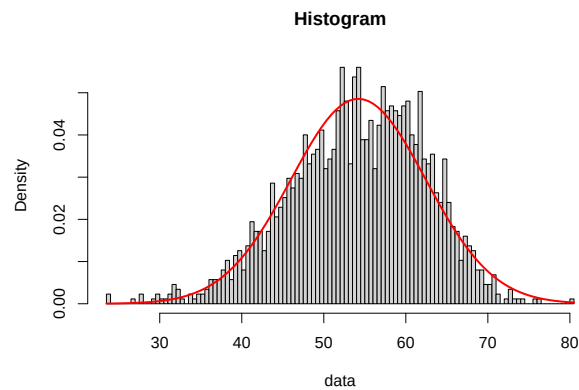
Variable *pctemployed16_over*

Numerical variable that represents the percentage of county residents ages 16 and over, employed.

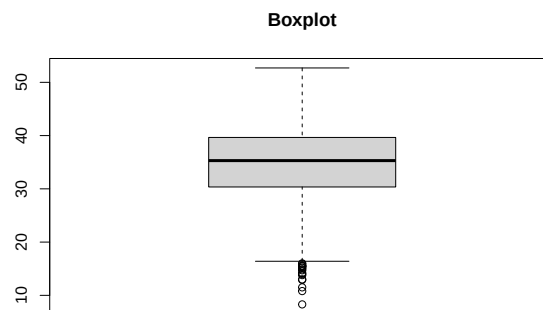
This variable contains over 82 NA values and has a slight right skewness and 11 mild outliers. Both inferential and visualization methods clearly point that the real distribution is extremely unlikely to be normal.

```
res.var <- summary_and_normality_check(train$pctemployed16_over)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's  
##  23.90  48.60   54.50   54.21  60.30   80.10    82  
## [1] "Mean"  
## [1] 54.21172  
## [1] "Standard deviation"  
## [1] 8.216153
```



```
## [1] "Skewness"  
## [1] -0.3362687
```



```
## [1] "Num Outliers:"
## [1] 11
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99196, p-value = 3.371e-08
```

```
outliers_df[res.var,"pctemployed16_over"] <- 1
```

Variable *pctunemployed16_over*

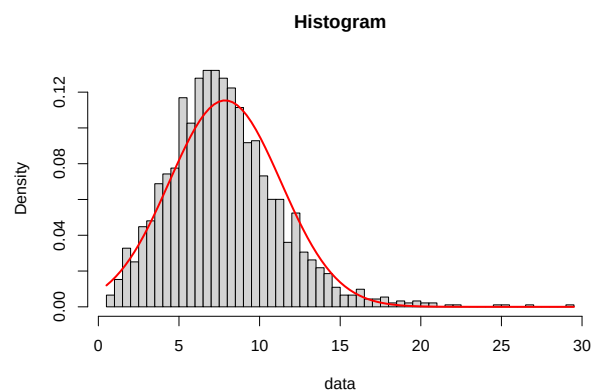
Numerical variable that represents the percentage of county residents ages 16 and over, unemployed.

This variable is heavily left skewed and contains over 46 outliers, of which 4 are severe. Despite the distribution looking almost normal there is a big number of outliers and a long heavy right tail. Applying a shapiro test we reject the hypothesis that the distribution is normal.

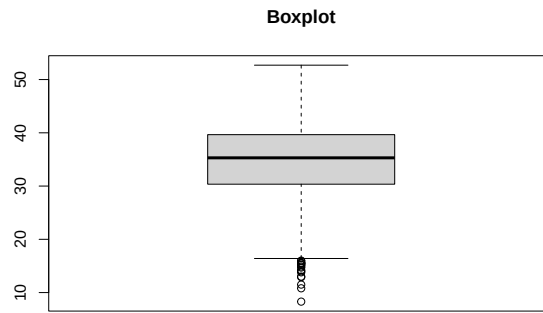
– op2: mild = 42 | sev = 4

```
res.var <- summary_and_normality_check(train$pctunemployed16_over)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.700  5.500   7.500   7.861  9.750  29.400
## [1] "Mean"
## [1] 7.86095
## [1] "Standard deviation"
## [1] 3.457111
```



```
## [1] "Skewness"
## [1] 0.9189968
```



```
## [1] "Num Outliers:"
## [1] 42
## [1] "Num. severe outliers:"
## [1] 4
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9612, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctunemployed16_over"] <- 1
```

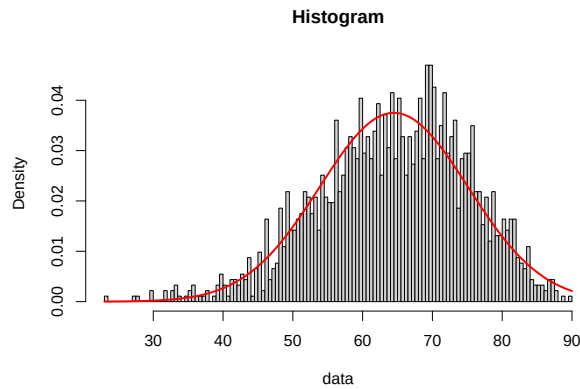
Variable *pctprivatecoverage*

Numerical variable that represents the percentage of county residents with private health coverage.

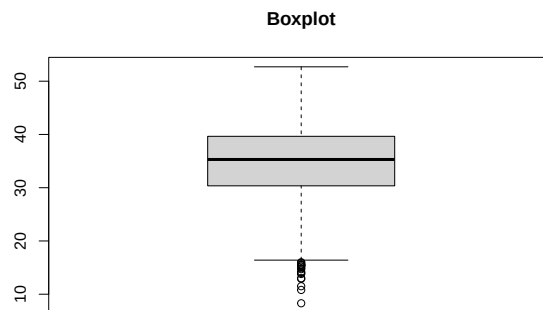
Despite having no NA and just 17 mild outliers, this variable has a huge standard deviation and a right skewed mean. Applying a shapiro test we reject the hypothesis that the distribution is normal.

```
res.var <- summary_and_normality_check(train$pctprivatecoverage)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  23.40  57.50   65.20   64.47  72.10   89.60
## [1] "Mean"
## [1] 64.4746
## [1] "Standard deviation"
## [1] 10.63614
```



```
## [1] "Skewness"
## [1] -0.3796678
```



```
## [1] "Num Outliers:"
## [1] 17
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98964, p-value = 3.725e-10
```

```
outliers_df[res.var,"pctprivatecoverage"] <- 1
```

Variable *pctprivatecoveragealone* We should check if the imputation is good enough to decide if we remove the variable or we keep it

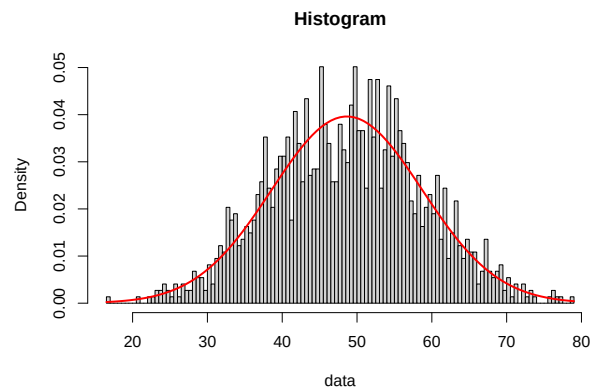
Numerical variable that represents the percentage of county residents with private health coverage alone (no public assistance at all).

The number of NA is humongous (356), but it has a small amount of mild outliers (4) and almost no skewness at all. Both using inferential and visualizations methods we cannot easily reject the hypothesis that this variable is not normal. So we can loosely say that is **normally** distributed.

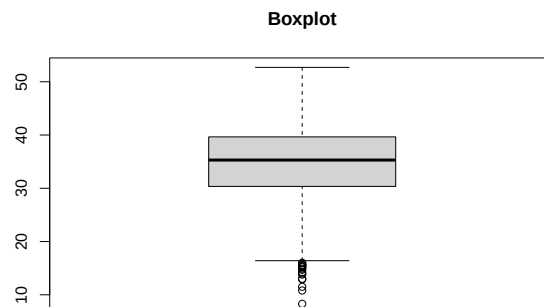
– op2 : After doing the Shapiro-Wilk test and obtaining a $p_value = 0,14$ we can not reject the null hypothesis, therefore this variable has a normal distribution.

```
res.var <- summary_and_normality_check(train$pctprivatecoveragealone)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      16.80  41.50   49.00   48.65  55.50   78.90   356
## [1] "Mean"
## [1] 48.64807
## [1] "Standard deviation"
## [1] 10.06977
```



```
## [1] "Skewness"
## [1] 0.01553065
```



```
## [1] "Num Outliers:"
## [1] 4
## [1] "Num. severe outliers:"
```

```
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99831, p-value = 0.1453
```

```
outliers_df[res.var,"pctprivatecoveragealone"] <- 1
```

Variable *pctempprivcoverage*

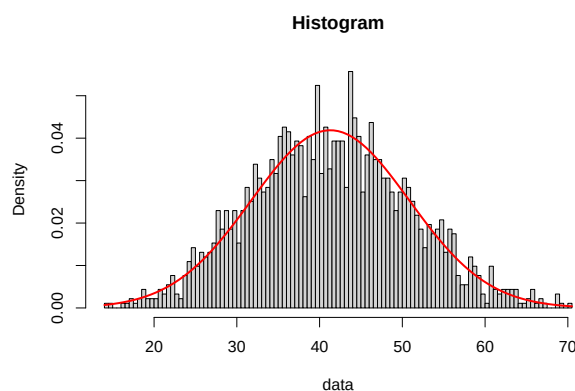
Numerical variable that represents the percentage of county residents with employee-provided private health coverage.

Despite the low number of outliers (7 mild), the absence of NA, the small left skewness and the median being almost equal to the mean, we should reject the null hypothesis that the distribution is normal with a 5% significance. However, with a smaller significance level than 3% we couldn't reject that this distribution is not normal.

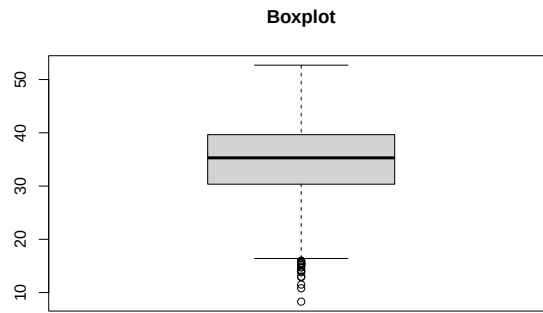
– we should clarify which is the C.level at the beginning of the project and then justify this part

```
res.var <- summary_and_normality_check(train$pctempprivcoverage)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  14.30  34.60   41.10   41.29  47.70   70.20
## [1] "Mean"
## [1] 41.29388
## [1] "Standard deviation"
## [1] 9.528869
```



```
## [1] "Skewness"
## [1] 0.1082829
```



```
## [1] "Num Outliers:"
## [1] 7
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99807, p-value = 0.02861
```

```
outliers_df[res.var,"pctempprivcoverage"] <- 1
```

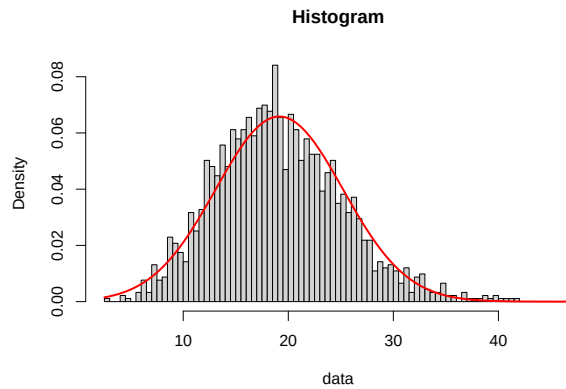
Variable *pctpubliccoveragealone*

Numerical variable that represents the percent of county residents with government-provided health coverage alone.

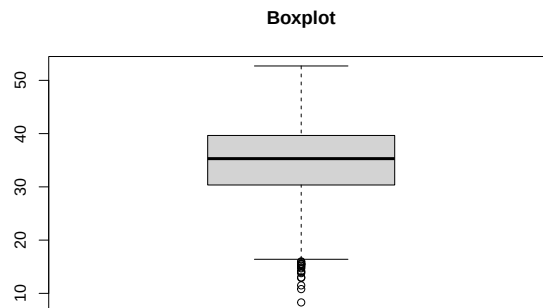
This variable contains 21 mild outliers, a great left skewness and a long right tail. Performing an inferential test like shapiro we get a tiny p-value which makes us reject the hypothesis that the distribution is normal under any significance level.

```
res.var <- summary_and_normality_check(train$pctpubliccoveragealone)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.60  14.90   18.70   19.15  23.00   46.60
## [1] "Mean"
## [1] 19.14992
## [1] "Standard deviation"
## [1] 6.056619
```



```
## [1] "Skewness"
## [1] 0.4695236
```



```
## [1] "Num Outliers:"
## [1] 21
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98784, p-value = 2.648e-11
```

```
outliers_df[res.var,"pctpubliccoveragealone"] <- 1
```

Variable *pctwhite*

Numerical variable that represents the percent of county residents who identify as White.

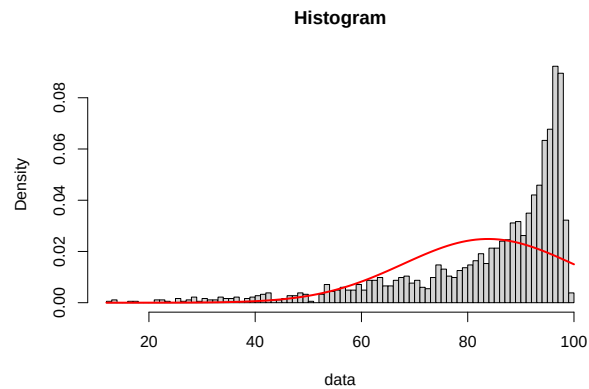
Both from the plot and from inferential methods we can easily discard that this variable is anywhere near normal, being a hugely right skewed variable. It contains 104 outliers from which 7 are severe.

This points out that the vast majority of counties have a high percentage of white population.

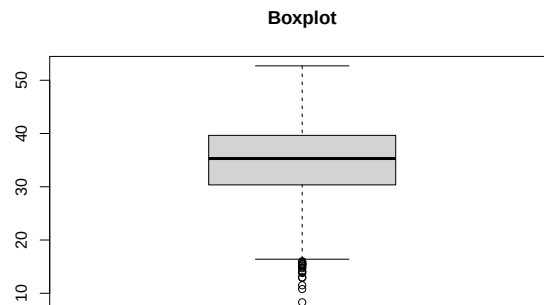
```
- op2 : mild = 97 | sev = 7
```

```
res.var <- summary_and_normality_check(train$pctwhite)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##    12.27  77.31   89.90   83.85  95.57   99.69  
## [1] "Mean"  
## [1] 83.84699  
## [1] "Standard deviation"  
## [1] 16.03708
```



```
## [1] "Skewness"  
## [1] -1.625604
```



```
## [1] "Num Outliers:"  
## [1] 97  
## [1] "Num. severe outliers:"  
## [1] 7  
## [1] "Shapiro test: "  
##  
##  Shapiro-Wilk normality test  
##  
## data:  data  
## W = 0.80758, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctwhite"] <- 1
```

Variable *pctblack*

Numerical variable that represents the percent of county residents who identify as Black.

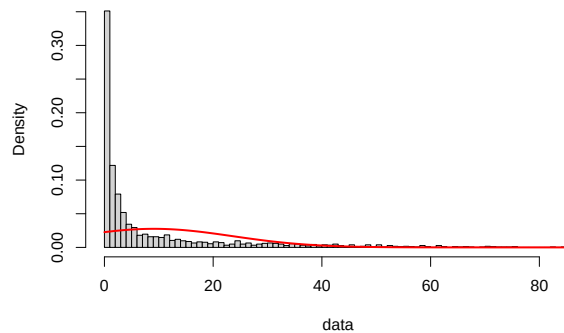
There is a huge number of outliers (323), from which 99 are severe and there is also a huge left skewness, that points out that the vast majority of counties (except the outliers) have a rather low percentage of black folks. The distribution almost looks like a power law distribution (Zipf) therefore we can safely reject both with visualization and inferential methods that the distribution of this variable is not normal.

– op2: mild = 224 | sev = 99

```
res.var <- summary_and_normality_check(train$pctblack)
```

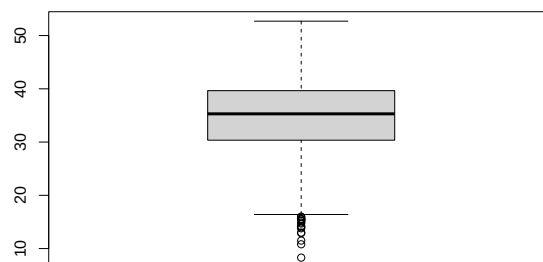
```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##  0.000  0.648   2.323   9.082 10.867  85.948  
## [1] "Mean"  
## [1] 9.081563  
## [1] "Standard deviation"  
## [1] 14.45309
```

Histogram



```
## [1] "Skewness"  
## [1] 2.272138
```

Boxplot



```
## [1] "Num Outliers:"
## [1] 224
## [1] "Num. severe outliers:"
## [1] 99
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.65926, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctblack"] <- 1
```

Variable *pctasian*

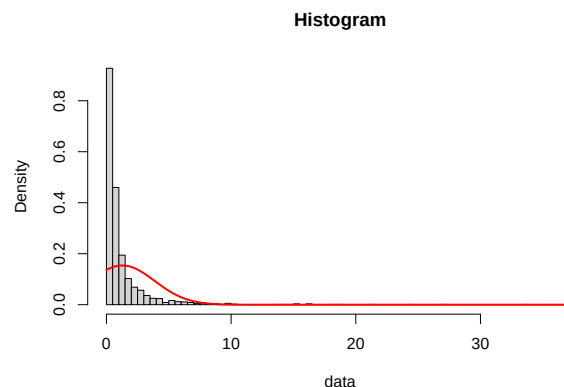
Numerical variable that represents the percent of county residents who identify as Asian.

Pretty similarly to the last distribution, we have a heavily left skewed zipfian like distribution, with 305 outliers (from which 107 are severe), pointing out that the vast majority of counties have less than 3% of asian percentage.

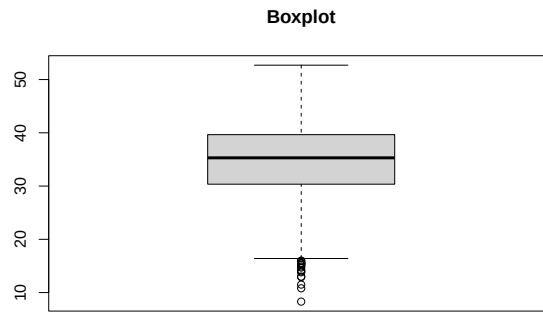
– op2: mild = 198 | sev = 107 – A: no acabo de entendre el resultat obtingut al boxplot dsp de mirar el histogarma

```
res.var <- summary_and_normality_check(train$pctasian)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.2582   0.5495   1.2743   1.2515  37.1569
## [1] "Mean"
## [1] 1.274301
## [1] "Standard deviation"
## [1] 2.588845
```

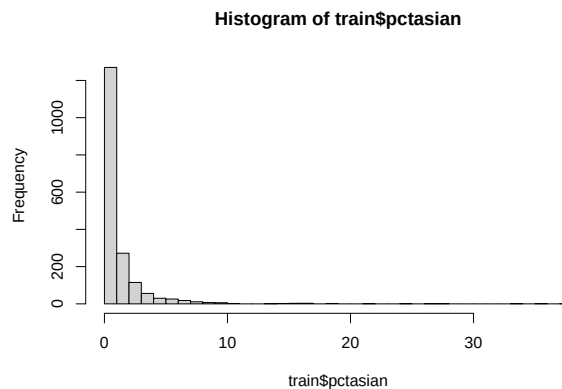


```
## [1] "Skewness"
## [1] 7.103922
```



```
## [1] "Num Outliers:"
## [1] 198
## [1] "Num. severe outliers:"
## [1] 107
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.41908, p-value < 2.2e-16
```

```
hist(train$pctasian , 50 )
```



```
outliers_df[res.var,"pctasian"] <- 1
```

Variable *pctotherrace*

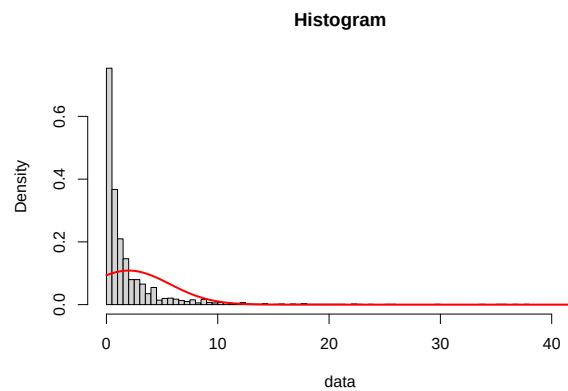
Numerical variable that represents the percent of county residents who identify as any other race (Pacific islanders, arábians, saiyajins, namekians...).

Again this variable, just like the last 2 presents a huge left skewness and zipfian like distribution with 283 outliers, from which 102 are severe. Again this sampled distribution is not normal.

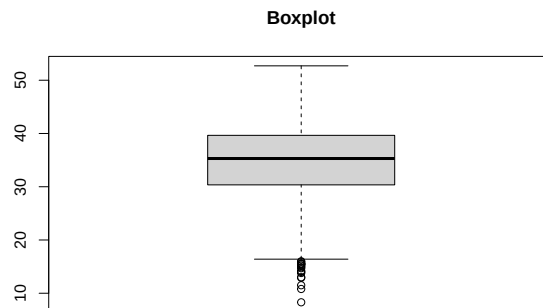
– op2 : mild = 181 | sev = 102

```
res.var <- summary_and_normality_check(train$pctotherrace)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
## 0.0000  0.2867  0.7826  2.0031  2.1066 41.9303  
## [1] "Mean"  
## [1] 2.003134  
## [1] "Standard deviation"  
## [1] 3.672809
```



```
## [1] "Skewness"  
## [1] 4.987083
```



```
## [1] "Num Outliers:"  
## [1] 181  
## [1] "Num. severe outliers:"  
## [1] 102  
## [1] "Shapiro test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.50981, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctotherrace"] <- 1
```

Variable *pctmarriedhouseholds*

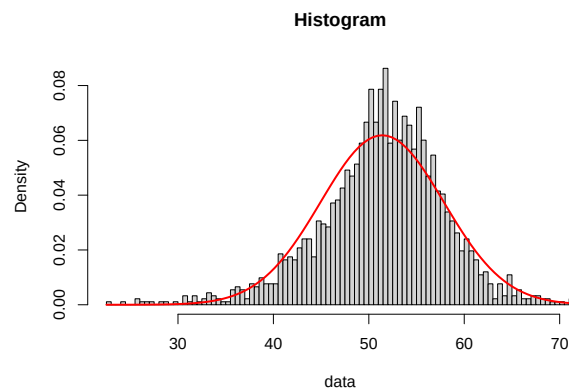
Numerical variable that represents the percent of married households.

This variable contains 59 outliers from which 2 are severe and a noticeable left skewness. Visualization isn't that clear, but using inferential methods like shapiro test we reject the null hypothesis that this variable is normal.

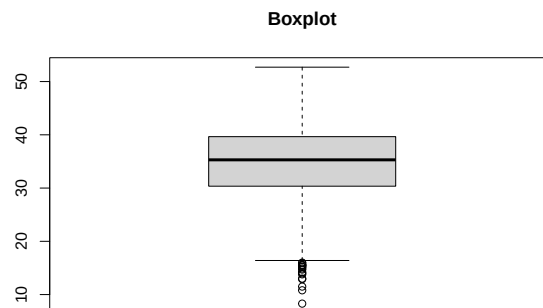
op2: mild = 57 | sev = 2

```
res.var <- summary_and_normality_check(train$pctmarriedhouseholds)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
## 22.99  47.85   51.73   51.40  55.48   71.40  
## [1] "Mean"  
## [1] 51.40232  
## [1] "Standard deviation"  
## [1] 6.450696
```



```
## [1] "Skewness"  
## [1] -0.5031018
```



```
## [1] "Num Outliers:"
## [1] 57
## [1] "Num. severe outliers:"
## [1] 2
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9816, p-value = 1.341e-14
```

```
outliers_df[res.var,"pctmarriedhouseholds"] <- 1
```

Variable *birthrate*

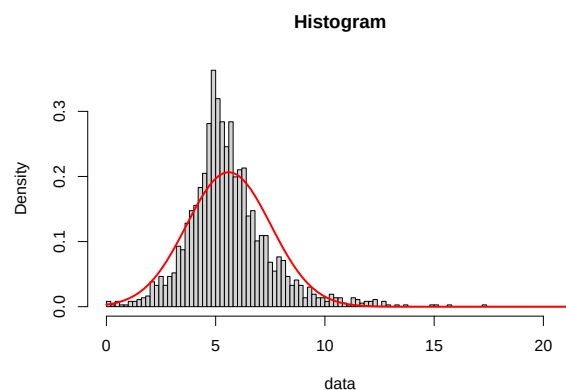
Numerical variable that represents the number of births relative to number of women in that specific county.

Finally, this last variable contains a fair amount of outliers (121) from which 17 are severe. It also has a fair amount of left skewness. Both from visualization and shapiro testing we reject that this sample variable is normal.

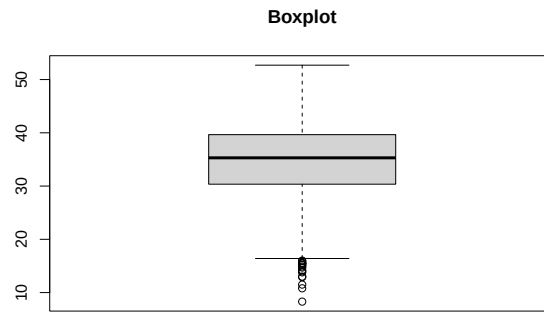
op2 : mild = 104 | sev = 17

```
res.var <- summary_and_normality_check(train$birthrate)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000  4.528   5.355   5.597   6.414   21.326
## [1] "Mean"
## [1] 5.597439
## [1] "Standard deviation"
## [1] 1.930891
```



```
## [1] "Skewness"
## [1] 1.246533
```



```
## [1] "Num Outliers:"
## [1] 104
## [1] "Num. severe outliers:"
## [1] 17
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.93107, p-value < 2.2e-16
```

```
outliers_df[res.var,"birthrate"] <- 1
```

Data Quality Report

Now that all the variables have been slightly analyzed in terms of their meaning. Given that univariate outliers and NA values have already been computed and shown, only a ranking and a visual summary will be showcased to gather the main insights visually.

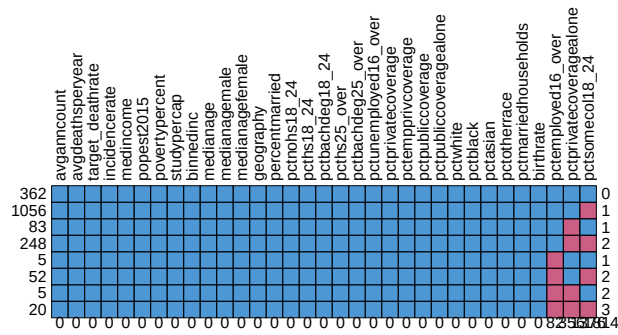
This quality report will be divided between variables and individuals of the dataset, and finally a multivariate outlier analysis will be performed.

Per variable

Per variable, count:

- Number of missing values.
- Number of errors (including inconsistencies).
- Number of outliers.
- Rank variables according to the sum of missing values (and errors).

```
# Number of errors (including inconsistencies). TODO
md.pattern(train, rotate.names=T)
```



##	avganncount	avgdeathspereyear	target_deathrate	incidence	rate	medincome
## 362	1	1	1	1	1	1
## 1056	1	1	1	1	1	1
## 83	1	1	1	1	1	1
## 248	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 52	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 20	1	1	1	1	1	1
##	0	0	0	0	0	0
##	popest2015	povertypercent	studypercap	binmedinc	medianage	medianagemale
## 362	1	1	1	1	1	1
## 1056	1	1	1	1	1	1
## 83	1	1	1	1	1	1
## 248	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 52	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 20	1	1	1	1	1	1
##	0	0	0	0	0	0
##	medianagefemale	geography	percentmarried	pctnohs18_24	pcths18_24	
## 362	1	1	1	1	1	
## 1056	1	1	1	1	1	
## 83	1	1	1	1	1	
## 248	1	1	1	1	1	
## 5	1	1	1	1	1	
## 52	1	1	1	1	1	
## 5	1	1	1	1	1	
## 20	1	1	1	1	1	
##	0	0	0	0	0	
##	pctbachdeg18_24	pcths25_over	pctbachdeg25_over	pctunemployed16_over		
## 362	1	1	1	1		
## 1056	1	1	1	1		
## 83	1	1	1	1		
## 248	1	1	1	1		
## 5	1	1	1	1		
## 52	1	1	1	1		
## 5	1	1	1	1		
## 20	1	1	1	1		

```

##          0          0          0          0
##      pctprivatecoverage pctempprivcoverage pctpubliccoverage
## 362          1          1          1          1
## 1056         1          1          1          1
## 83          1          1          1          1
## 248         1          1          1          1
## 5           1          1          1          1
## 52          1          1          1          1
## 5           1          1          1          1
## 20          1          1          1          1
##          0          0          0          0
##      pctpubliccoveragealone pctwhite pctblack pctasian pctotherrace
## 362          1          1          1          1          1
## 1056         1          1          1          1          1
## 83          1          1          1          1          1
## 248         1          1          1          1          1
## 5           1          1          1          1          1
## 52          1          1          1          1          1
## 5           1          1          1          1          1
## 20          1          1          1          1          1
##          0          0          0          0          0
##      pctmarriedhouseholds birthrate pctemployed16_over pctprivatecoveragealone
## 362          1          1          1          1
## 1056         1          1          1          1
## 83          1          1          1          0
## 248         1          1          1          0
## 5           1          1          0          1
## 52          1          1          0          1
## 5           1          1          0          0
## 20          1          1          0          0
##          0          0          82          356
##      pctsomecol18_24
## 362          1  0
## 1056         0  1
## 83          1  1
## 248         0  2
## 5           1  1
## 52          0  2
## 5           1  2
## 20          0  3
##          1376 1814

```

On one hand, looking at the pattern, we can notice that there are 362 variables with no null values, and that the vast majority of nulls (1056) are located on the pctsomecol18_24 variable with just 1 null column. Another remarkable trend is that up to 248 individuals have 2 null values in variables pctsomecol18_24 and pctprivatecoveragealone, while this other variable just has 83 pure nulls (Only this column). The last variable that contains a low amount of null values is pctemployed16_over, where 20 individuals have the last 3 mentioned variables containing nulls, and the rest of few possible NA values come from pattern permutations of these 3 variables. Therefore this dataset only has 3 possible rows containing NA, and only one of them is relevant in this aspect. Ranking:

Rank	Null values
1	pctsomecol18_24: 1375 NA

Rank	Null values
2	pctprivatecoveragealone: 356 NA
4	avganncount : 126 NA
4	incidencerate : 126 NA
3	pctemployed16_over: 82 NA

On the other hand, the top 5 ranking for variables with the most amount of mild and severe outliers is:

TO_DO -> CHECK

Rank	Mild Outliers	Severe Outliers
1	avganncount 333	avganncount 257
2	pctblack 323	studypercap 222
3	studypercap 307	popest2015 162
4	pctasian 305	avgdeathsperyear 138
5	pctotherrace 283	pctasian 107

Per individuals

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariant outliers.

There are 362 obs without NA in any variable, 1144 obs with 1 variable with a NA value, 305 obs with 2 NA values and 20 obs with 3 NA values along the 33 variables of the dataset.

```
#Number of missing values
na_counts <- rowSums(is.na(train))
table(na_counts)
```

```
## na_counts
##    0    1    2    3
## 362 1144 305  20
```

Out of all observations, 971 do not exhibit any outliers across the variables, while 385 observations contain a single outlier in one variable. Interestingly, there is 1 observation where we identified outliers in 10 variables and another 10 observations that are outliers in 8 variables.

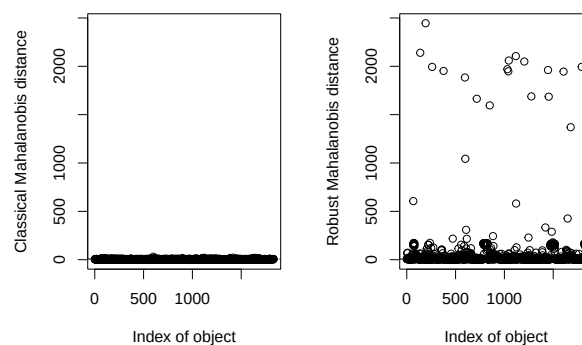
```
#Number of outliers per individual
outliers_counts <- rowSums(outliers_df)
table(outliers_counts)
```

```
## outliers_counts
##    0    1    2    3    4    5    6    7    8    9   10   11
## 846 443 162 114  96  70  51  19  19   7   2   2
```

Considering over 20 numerical variables that contain few correlated variables and considering a 1% significance level, 209 multivariate outliers are found in the dataset.

```
#Identify individuals considered as multivariant outliers.
train$avgdeathspereyear <- as.numeric(train$avgdeathspereyear)
train$medincome <- as.numeric(train$medincome)
train$popest2015 <- as.numeric(train$popest2015)

#Compute multivariant outliers and find their ids (24)
subset_num_vars <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
res_mout <- Moutlier(train[,subset_num_vars], quantile=0.97, plot=T)
```



```
mout_ids <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff))

#Reset graphical parameters and plot the outliers
dev.off()
```

```
## null device
##          1
```

```
plot(x=res_mout$md, y=res_mout$rd, main="Classic Mahalanobis D vs Robust Mahalanobis D (M. Outliers)",
abline(v=res_mout$cutoff, col="red")
abline(h=res_mout$cutoff, col="red")
```

We can clearly see that for the number of outliers variable, the mean is around 1.39, and the third quantile around 2.0, however, we have a high standard deviation of 1.935 pointing out at the presence of outliers in the number of outliers variable, suggesting there are several individuals with more than 2 outlier variables. This variable contains somewhat high correlation with pths25_over, pchwhite, pctasian and other weak correlations.

The number of missing values doesn't make much sense to be analyzed after removing 2 of the main problematic columns which contained several null values.

Finally, looking at the multivariate factor profiling, we see that there is a global association between "mout" and "binnedinc", and we can see that there is a huge global association with quantitative variables pctoterrace and n_uout (Logically, the number of univariate outliers contributes to a possible multivariate outlier individual), followed by medianage looking at their eta squared values. Finally, we see looking at the mean values of the quantitative variables that in the case of an individual that is not a multivariate outlier, almost every single variable mean value is equal or close to the overall mean. However in the case

of multivariate outliers we see huge discrepancies in the mean of pctasian, pctotherrace, n_out, popest2015, avgdeathspereyear and avganncount, which are way higher than the overall mean value. This may point out that the asian-otherrace populations and those with higher population or more average deaths per year, are more prone to be multivariate outliers.

```
anal_df <- train
#Binary factor if its a multivariate outlier
anal_df$mout <- 0
anal_df[mout_ids, "mout"] <- 1
anal_df$mout <- factor(anal_df$mout, labels=c("Not M.O", "Yes M.O"))
#Number of univariate outlier variables for a given observation
anal_df$n_uout <- rowSums(outliers_df)
#Number of missing values in variables for a given observation
anal_df$n_miss <- rowSums(is.na(train))

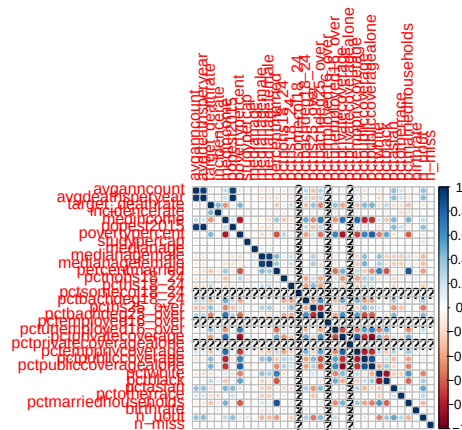
summary(anal_df[, c("mout", "n_uout", "n_miss")]); sd(anal_df$n_uout); sd(anal_df$n_miss)
```

##	mout	n_uout	n_miss
##	Not M.O:1606	Min. : 0.000	Min. :0.0000
##	Yes M.O: 225	1st Qu.: 0.000	1st Qu.:1.0000
##		Median : 1.000	Median :1.0000
##		Mean : 1.387	Mean :0.9907
##		3rd Qu.: 2.000	3rd Qu.:1.0000
##		Max. :11.000	Max. :3.0000

```
## [1] 1.938147
```

```
## [1] 0.6388352
```

```
num_vars <- anal_df[sapply(anal_df, is.numeric)]
corrplot(cor(num_vars))
```



```
catdes(anal_df, num.var=34, proba=0.05)
```

```
##
## Link between the cluster variable and the categorical variables (chi-square test)
## =====
```

```

##                p.value df
## binnedinc 0.0001182852  9
##
## Description of each cluster by the categories
## =====
## $'Not M.O'
##                Cla/Mod  Mod/Cla  Global  p.value
## binnedinc=(51046.4, 54545.6] 93.75000 10.273973 9.612234 0.0064710699
## binnedinc=(48021.6, 51046.4] 92.30769 11.207970 10.649918 0.0320467887
## binnedinc=[22640, 34218.1] 80.41237 9.713574 10.595303 0.0020447042
## binnedinc=(61494.5, 125635] 79.56989 9.215442 10.158383 0.0008458263
##                v.test
## binnedinc=(51046.4, 54545.6] 2.722905
## binnedinc=(48021.6, 51046.4] 2.143827
## binnedinc=[22640, 34218.1] -3.083661
## binnedinc=(61494.5, 125635] -3.337346
##
## $'Yes M.O'
##                Cla/Mod  Mod/Cla  Global  p.value
## binnedinc=(61494.5, 125635] 20.430108 16.888889 10.158383 0.0008458263
## binnedinc=[22640, 34218.1] 19.587629 16.888889 10.595303 0.0020447042
## binnedinc=(48021.6, 51046.4] 7.692308 6.666667 10.649918 0.0320467887
## binnedinc=(51046.4, 54545.6] 6.250000 4.888889 9.612234 0.0064710699
##                v.test
## binnedinc=(61494.5, 125635] 3.337346
## binnedinc=[22640, 34218.1] 3.083661
## binnedinc=(48021.6, 51046.4] -2.143827
## binnedinc=(51046.4, 54545.6] -2.722905
##
##
## Link between the cluster variable and the quantitative variables
## =====
##                Eta2      P-value
## n_uout          0.145966349 1.049888e-64
## pctotherrace    0.099628021 1.225624e-43
## medianage       0.062554722 1.635718e-27
## pctwhite        0.059116765 4.782414e-26
## pctasian        0.045841171 1.988142e-20
## pcths25_over    0.043595539 1.748772e-19
## popest2015      0.034525185 1.098770e-15
## avganncount     0.032527270 7.489993e-15
## avgdeathsperyear 0.030126599 7.496950e-14
## pctempprivcoverage 0.019328031 2.316662e-09
## percentmarried  0.019175132 2.681751e-09
## incidencerate   0.016324389 4.112895e-08
## pctnohs18_24    0.015285419 1.113888e-07
## pctprivatecoverage 0.013413434 6.724710e-07
## povertypercent  0.013283435 7.620289e-07
## target_deathrate 0.011450565 4.455514e-06
## medianagefemale 0.010053541 1.720632e-05
## medianagemale   0.009396851 3.253972e-05
## birthrate       0.009256862 3.728142e-05
## pctbachdeg25_over 0.009091283 4.379394e-05
## pctbachdeg18_24 0.008074303 1.180369e-04

```

```

## pctmarriedhouseholds    0.007426363 2.226259e-04
## pctunemployed16_over    0.005856003 1.048909e-03
## pctemployed16_over      0.004120889 7.241760e-03
## pctprivatecoveragealone 0.004498010 9.981131e-03
## pctsomecol18_24        0.014549394 1.001640e-02
## medincome               0.003262597 1.450668e-02
##
## Description of each cluster by quantitative variables
## =====
## $'Not M.O'
##
##          v.test Mean in category Overall mean sd in category
## pctwhite      10.401138      85.306077 8.384699e+01 1.435103e+01
## pcths25_over   8.931956      35.284620 3.473255e+01 6.715734e+00
## pctempprivcoverage 5.947293      41.789601 4.129388e+01 8.979987e+00
## percentmarried 5.923723      52.250125 5.190087e+01 6.086924e+00
## incidencerate  5.465678     451.661576 4.489919e+02 4.730818e+01
## pctsomecol18_24 5.372873      40.996456 4.047846e+01 1.048402e+01
## pctprivatecoverage 4.954451      64.935554 6.447460e+01 9.772785e+00
## target_deathrate 4.577612     179.938045 1.788197e+02 2.528725e+01
## medianagefemale 4.289287      42.364259 4.216931e+01 4.743624e+00
## medianagemale  4.146835      39.773848 3.958564e+01 4.733155e+00
## pctmarriedhouseholds 3.686495      51.610338 5.140232e+01 5.883973e+00
## pctprivatecoveragealone 2.893764      48.902943 4.864807e+01 9.470267e+00
## pctemployed16_over 2.724793      54.407547 5.421172e+01 7.621998e+00
## medincome      -2.443472     47014.155044 4.727794e+04 1.115646e+04
## pctunemployed16_over -3.273604      7.761955 7.860950e+00 3.096049e+00
## pctbachdeg18_24 -3.843953      6.064321 6.216221e+00 3.813470e+00
## pctbachdeg25_over -4.078854      13.102366 1.329754e+01 5.103788e+00
## birthrate      -4.115830      5.527922 5.597439e+00 1.694128e+00
## povertypercent  -4.930384      16.515255 1.679175e+01 5.851570e+00
## pctnohs18_24   -5.288886      17.921357 1.828777e+01 7.111371e+00
## avgdeathsperyear -7.425071      154.480697 1.916237e+02 2.701091e+02
## avganncount     -7.715238      516.055275 6.232245e+02 8.349126e+02
## popest2015      -7.948653      80663.803861 1.068412e+05 1.580656e+05
## pctasian        -9.159113      1.066889 1.274301e+00 1.550148e+00
## medianage       -10.699306      41.051868 4.524970e+01 4.701650e+00
## pctotherrace    -13.502566      1.569335 2.003134e+00 2.271112e+00
## n_uout          -16.343758      1.109589 1.386674e+00 1.644780e+00
##
##          Overall sd      p.value
## pctwhite      1.603270e+01 2.449885e-25
## pcths25_over   7.064060e+00 4.185422e-19
## pctempprivcoverage 9.526266e+00 2.726128e-09
## percentmarried 6.738289e+00 3.147337e-09
## incidencerate  5.582455e+01 4.611418e-08
## pctsomecol18_24 1.101857e+01 7.749189e-08
## pctprivatecoverage 1.063324e+01 7.253484e-07
## target_deathrate 2.792279e+01 4.703134e-06
## medianagefemale 5.194588e+00 1.792479e-05
## medianagemale  5.187250e+00 3.371034e-05
## pctmarriedhouseholds 6.448935e+00 2.273640e-04
## pctprivatecoveragealone 1.006636e+01 3.806547e-03
## pctemployed16_over 8.213804e+00 6.434182e-03
## medincome      1.233832e+04 1.454672e-02
## pctunemployed16_over 3.456167e+00 1.061851e-03

```

```

## pctbachdeg18_24      4.516322e+00 1.210682e-04
## pctbachdeg25_over    5.468847e+00 4.525832e-05
## birthrate            1.930363e+00 3.857885e-05
## povertypercent       6.409410e+00 8.206812e-07
## pctnohs18_24         7.917877e+00 1.230634e-07
## avgdeathsperyear     5.717202e+02 1.127197e-13
## avganncount          1.587552e+03 1.207562e-14
## popest2015           3.763912e+05 1.885499e-15
## pctasian             2.588138e+00 5.232582e-20
## medianage            4.484112e+01 1.025437e-26
## pctotherrace         3.671806e+00 1.510237e-41
## n_uout               1.937617e+00 4.818920e-60
##
## $'Yes M.O'
##
##          v.test Mean in category Overall mean sd in category
## n_uout      16.343758      3.364444e+00 1.386674e+00 2.604369e+00
## pctotherrace 13.502566      5.099498e+00 2.003134e+00 7.871966e+00
## medianage    10.699306      7.521289e+01 4.524970e+01 1.232132e+02
## pctasian      9.159113      2.754761e+00 1.274301e+00 5.904230e+00
## popest2015    7.948653      2.936894e+05 1.068412e+05 9.668209e+05
## avganncount    7.715238      1.388175e+03 6.232245e+02 3.855787e+03
## avgdeathsperyear 7.425071      4.567422e+02 1.916237e+02 1.434938e+03
## pctnohs18_24   5.288886      2.090311e+01 1.828777e+01 1.189171e+01
## povertypercent 4.930384      1.876533e+01 1.679175e+01 9.244450e+00
## birthrate      4.115830      6.093634e+00 5.597439e+00 3.091473e+00
## pctbachdeg25_over 4.078854      1.469067e+01 1.329754e+01 7.432688e+00
## pctbachdeg18_24 3.843953      7.300444e+00 6.216221e+00 7.800373e+00
## pctunemployed16_over 3.273604      8.567556e+00 7.860950e+00 5.312055e+00
## medincome       2.443472      4.916081e+04 4.727794e+04 1.861157e+04
## pctemployed16_over -2.767633      5.279198e+01 5.421172e+01 1.153767e+01
## pctprivatecoveragealone -2.844516      4.685978e+01 4.864807e+01 1.339354e+01
## pctmarriedhouseholds -3.686495      4.991756e+01 5.140232e+01 9.423835e+00
## medianagemale    -4.146835      3.824222e+01 3.958564e+01 7.550120e+00
## medianagefemale  -4.289287      4.077778e+01 4.216931e+01 7.534321e+00
## target_deathrate -4.577612      1.708369e+02 1.788197e+02 4.132807e+01
## pctprivatecoverage -4.954451      6.118444e+01 6.447460e+01 1.503502e+01
## pctsomecol18_24  -4.955522      3.706833e+01 4.047846e+01 1.355299e+01
## incidencerate    -5.465678      4.299362e+02 4.489919e+02 9.471860e+01
## percentmarried   -5.923723      4.940800e+01 5.190087e+01 9.896851e+00
## pctempprivcoverage -5.947293      3.775556e+01 4.129388e+01 1.219163e+01
## pcths25_over     -8.931956      3.079200e+01 3.473255e+01 8.152171e+00
## pctwhite        -10.401138      7.343237e+01 8.384699e+01 2.231803e+01
##
##          Overall sd      p.value
## n_uout      1.937617e+00 4.818920e-60
## pctotherrace 3.671806e+00 1.510237e-41
## medianage    4.484112e+01 1.025437e-26
## pctasian      2.588138e+00 5.232582e-20
## popest2015    3.763912e+05 1.885499e-15
## avganncount    1.587552e+03 1.207562e-14
## avgdeathsperyear 5.717202e+02 1.127197e-13
## pctnohs18_24   7.917877e+00 1.230634e-07
## povertypercent 6.409410e+00 8.206812e-07
## birthrate      1.930363e+00 3.857885e-05
## pctbachdeg25_over 5.468847e+00 4.525832e-05

```

```
## pctbachdeg18_24      4.516322e+00 1.210682e-04
## pctunemployed16_over  3.456167e+00 1.061851e-03
## medincome            1.233832e+04 1.454672e-02
## pctemployed16_over   8.213804e+00 5.646501e-03
## pctprivatecoveragealone 1.006636e+01 4.447890e-03
## pctmarriedhouseholds 6.448935e+00 2.273640e-04
## medianagemale        5.187250e+00 3.371034e-05
## medianagefemale      5.194588e+00 1.792479e-05
## target_deathrate     2.792279e+01 4.703134e-06
## pctprivatecoverage    1.063324e+01 7.253484e-07
## pctsomecol18_24      1.101857e+01 7.213639e-07
## incidencerate        5.582455e+01 4.611418e-08
## percentmarried       6.738289e+00 3.147337e-09
## pctempprivcoverage    9.526266e+00 2.726128e-09
## pcths25_over         7.064060e+00 4.185422e-19
## pctwhite             1.603270e+01 2.449885e-25
```

Imputation

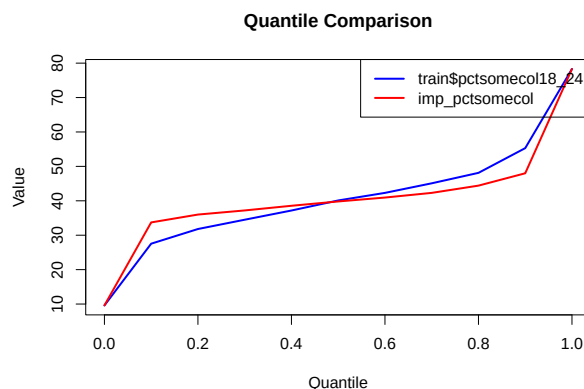
pctsomecol18_24

Given the huge amount of null values, the impute PCA has not been able to replicate the original distribution of this messy variable.

```
num_vars <- train[sapply(train, is.numeric)]
miss_ids <- which(is.na(train$pctsomecol18_24))
res_imp <- imputePCA(num_vars)
imp_pctsomecol <- res_imp$completeObs[, "pctsomecol18_24"]

# Calculate the quantiles for validation
quantiles_train <- quantile(train$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctsomecol, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
     xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("train$pctsomecol18_24", "imp_pctsomecol"),
     col = c("blue", "red"), lwd = 2)
```



```
#We won't impute the values!
train[miss_ids, "pctsomecol18_24"] <- imp_pctsomecol[miss_ids]
```

```
# --- op2 ---
summary(train$pctsomecol18_24)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      9.60  36.60   39.81   40.33   43.17   78.30
```

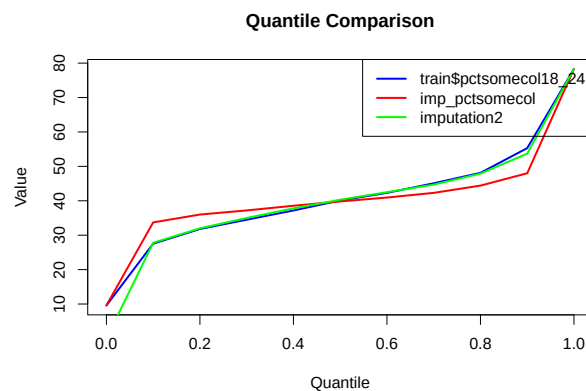
```
imp_values <- 100 - train$pctnohs18_24[miss_ids] - train$pcths18_24[miss_ids] - train$pctbachdeg18_24[m
pct2 <- train$pctsomecol18_24
pct2[miss_ids] <- imp_values
```

```
quantiles_2 <- quantile(pct2, probs = seq(0,1,0.1), na.rm = FALSE)
summary(pct2)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00  33.45   40.30   40.50   46.05   78.30
```

```
plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
     xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
```

```
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2);
lines(seq(0, 1, 0.1), quantiles_2, col = "green", lwd = 2);
legend("topright", legend = c("train$pctsomecol18_24", "imp_pctsomecol", "imputation2"),
     col = c("blue", "red", "green"), lwd = 2)
```



```
# add imputed values
train$pctsomecol18_24 <- pct2
```

pctemployed16_over

pctemployed16_over only contains 70 null values and the imputePCA function has been able of almost perfectly preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

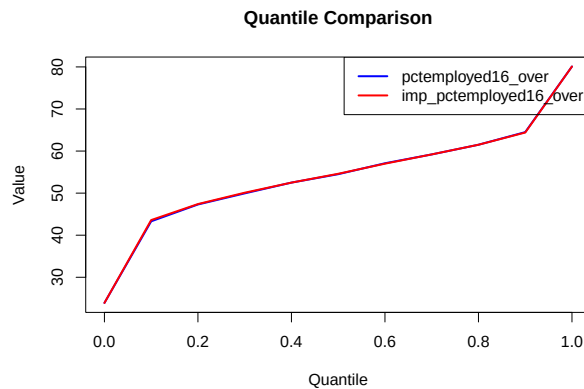
```

num_vars <- train[sapply(train, is.numeric)]
miss_ids <- which(is.na(train$pctemployed16_over))
res_imp <- imputePCA(num_vars)
imp_pctemployed16_over <- res_imp$completeObs[, "pctemployed16_over"]

# Calculate the quantiles for validation
quantiles_train <- quantile(train$pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("pctemployed16_over", "imp_pctemployed16_over"),
      col = c("blue", "red"), lwd = 2)

```



```

train[miss_ids, "pctemployed16_over"] <- imp_pctemployed16_over[miss_ids]

```

pctprivatecoveragealone

Similarly, the variable pctprivatecoveragealone is better informed despite a quite high number of null values (253) and the imputePCA has been successful to preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

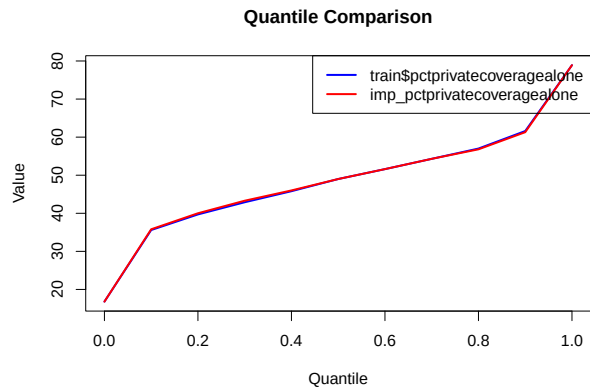
```

num_vars <- train[sapply(train, is.numeric)]
miss_ids <- which(is.na(train$pctprivatecoveragealone))
res_imp <- imputePCA(num_vars)
imp_pctprivatecoveragealone <- res_imp$completeObs[, "pctprivatecoveragealone"]

# Calculate the quantiles for validation
quantiles_train <- quantile(train$pctprivatecoveragealone, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctprivatecoveragealone, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("train$pctprivatecoveragealone", "imp_pctprivatecoveragealone"),
      col = c("blue", "red"), lwd = 2)

```



```
train[miss_ids, "pctprivatecoveragealone"] <- imp_pctprivatecoveragealone[miss_ids]
```

Profiling & Relationships

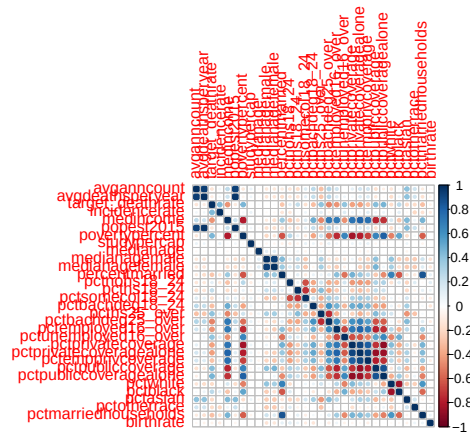
After imputation and removal of null values, we can proceed with profiling the target variable and assess if any variable contains serial correlation.

TODO: explicar correlacions... (Cuidado, no cal explicar les correlacions de target_deathrate, ja estan just al apartat d'abaix!!)

```
names(train)
```

```
## [1] "avganncount"          "avgdeathspereyear"
## [3] "target_deathrate"    "incidencerate"
## [5] "medincome"           "popest2015"
## [7] "povertypercent"      "studypercap"
## [9] "binnedinc"           "medianage"
## [11] "medianagemale"       "medianagefemale"
## [13] "geography"           "percentmarried"
## [15] "pctnohs18_24"        "pcths18_24"
## [17] "pctsomecol18_24"     "pctbachdeg18_24"
## [19] "pcths25_over"        "pctbachdeg25_over"
## [21] "pctemployed16_over"  "pctunemployed16_over"
## [23] "pctprivatecoverage"  "pctprivatecoveragealone"
## [25] "pctempprivcoverage"  "pctpubliccoverage"
## [27] "pctpubliccoveragealone" "pctwhite"
## [29] "pctblack"            "pctasian"
## [31] "pctotherrace"        "pctmarriedhouseholds"
## [33] "birthrate"
```

```
num_vars <- train[sapply(train, is.numeric)]
cat_vars <- train[!sapply(train, is.numeric)]
corrplot(cor(num_vars))
```



For some unexpected reason the geography categorical variable causes some unexpected problems to arise, therefore we won't use it at all. If you take a closer look at the profiling of this target variable you will realize that there is a high linkage between the target and the qualitative variable binnedinc performing a oneway anova test. If you take a look at the mean values for different values of binnedinc, you will see that those incomes which are below 40K have a higher deathrate (The lower the binned income the higher the deathrate). Conversely, the higher the binned income the lower the deathrate. Finally, taking a glance at the quantitative variables, we see that there is a medium-high positive correlation with pctpubliccoveragealone, povertyperecent, pctpubliccoverage, pcths25_over and pctunemployed16_over. Looking at the inverse, we see a medium-high inverse correlation with variables medincome, pctbachdeg25_over and pctprivatecoverage and pctmarriedhouseholds.

```
subset_num <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
subset_cat <- c(9)
subset_train <- c(subset_num, subset_cat)

res_anal <- condres(train[, subset_train], num.var=3, proba=0.01)
res_anal
```

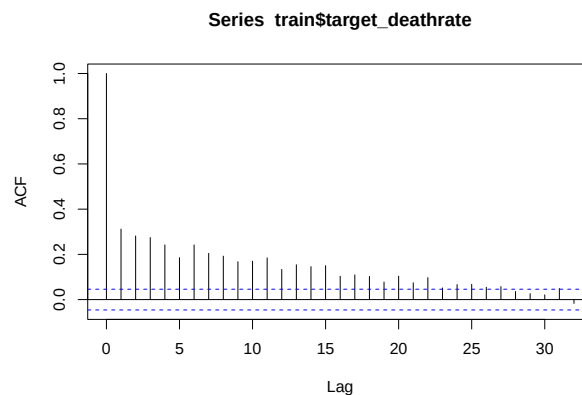
```
##
## Link between the variable and the continuous variables (R-square)
## =====
## correlation p.value
## pctpubliccoveragealone 0.46629578 1.687857e-99
## povertyperecent 0.45008465 5.144264e-92
## incidencerate 0.44954847 8.953861e-92
## pctpubliccoverage 0.42436751 6.056303e-81
## pcths25_over 0.40690172 5.910723e-74
## pctunemployed16_over 0.39780681 1.777035e-70
## pctblack 0.26875452 1.156196e-31
## pcths18_24 0.24243487 6.687515e-26
## pctnohs18_24 0.11480060 8.424149e-07
## birthrate -0.06756043 3.824884e-03
## avgdeathsperyear -0.09167650 8.549496e-05
## popest2015 -0.11902232 3.261480e-07
## avganncount -0.13886454 2.417857e-09
## pctwhite -0.18152553 4.983753e-15
## pctotherrace -0.18153593 4.965708e-15
## pctasian -0.19436412 4.806194e-17
## percentmarried -0.27253733 1.511951e-32
```

```
## pctempprivcoverage      -0.28877690  1.677032e-36
## pctmarriedhouseholds    -0.29440643  6.182702e-38
## pctbachdeg18_24         -0.30162255  8.054861e-40
## pctprivatecoverage      -0.40354149  1.173008e-72
## medincome               -0.44368280  3.606607e-89
## pctbachdeg25_over       -0.49777156  3.106333e-115
##
## Link between the variable and the categorical variable (1-way anova)
## =====
##              R2      p.value
## binnedinc 0.2217324 7.669702e-93
##
## Link between variable and the categories of the categorical variables
## =====
##              Estimate      p.value
## binnedinc=[22640, 34218.1] 25.763711 9.853130e-45
## binnedinc=(34218.1, 37413.8] 13.511997 1.196419e-11
## binnedinc=(37413.8, 40362.7]  9.454553 3.683926e-06
## binnedinc=(48021.6, 51046.4] -7.952886 3.574891e-05
## binnedinc=(51046.4, 54545.6] -9.460955 3.130385e-06
## binnedinc=(54545.6, 61494.5] -13.093413 1.157068e-11
## binnedinc=(61494.5, 125635] -20.458908 2.459391e-26
```

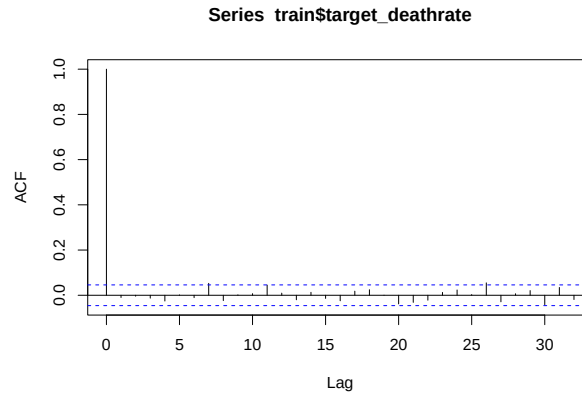
Serial correlations

It has been found previously that the target_variable doesn't follow a normal distribution at all, now its clear looking at the auto correlation function plot below that this variable contains serial correlation. To address this issue we will

```
library(stats)
acf(train$target_deathrate)
```



```
ids <- sample(1:nrow(train), nrow(train))
train <- train[ids, ]
acf(train$target_deathrate)
```



Modeling (train data)

Initial Model exploration and Multicollinearity study

Firstly, it is important to realize that just as in past sections, the geography variable should just be used as a supplementary variable and not used for modelling purposes. Two initial models are built, one with all the numeric variables, and a second one containing all the numeric variables + all categorical variables.

Both models explain roughly 51% and 52% of variance respectively (R^2), and both have multiple significant slope coefficients such as percentmarried, pctbachdeg25_over, pctemployed16_over, pctotherrace, avganncount, incidencerate, avgdeathspereyear and binnedinc level 7 among the ones with smallest p-values.

Looking at variance over inflation of the first model, we can see that there are several variables that contain a great amount of variance over inflation, pointing out the presence of multicollinearity. The main ranked variables that exceed the high VIF threshold (10) in both models:

1. avgdeathspereyear
2. pctpubliccoverage
3. popest2015
4. pctprivatecoveragealone
5. pctprivatecoverage
6. pctpubliccooveragealone
7. avganncount
8. percentmarried
9. binnedinc

Exploring the residual analysis plots of each model, a common trend is easily identified. Due to the presence of extreme outliers, both models do not meet the linear model premises at all:

1. Residuals vs Fitted plots look like residuals are random and do not follow any pattern except for a couple of extreme outliers, which distort the straight line, therefore we cannot claim linearity.
2. Q-Q Residuals suggest that the residuals almost follow a normal distribution (are random), despite a few outlier and heavier tails than usual, therefore we cannot fully claim normality.
3. Scale-Location plots show a slight curvature due to the outlier values that points out a possible small heteroscedasticity.
4. Residuals vs Leverage confirms that despite almost all points have a quite low leverage, there are a small amount of highly leveraged points that distort the model in general (Specially observation 600).

```
m1 <- lm(target_deathrate ~ avganncount + avgdeathsperyear + incidencerate + medincome + popest2015 + p
summary(m1)
```

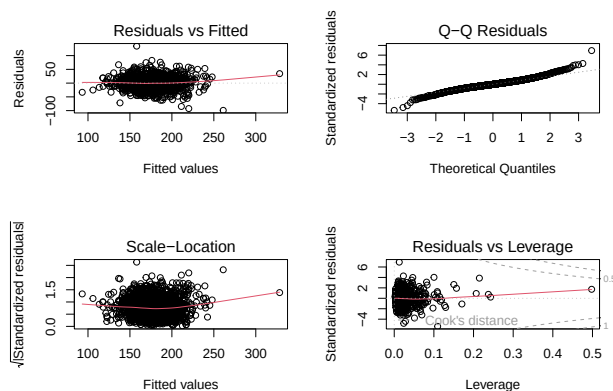
```
##
## Call:
## lm(formula = target_deathrate ~ avganncount + avgdeathsperyear +
##     incidencerate + medincome + popest2015 + povertypercent +
##     studypercap + medianage + medianagefemale + medianagemale +
##     percentmarried + pctnohs18_24 + pcths18_24 + pctsomecol18_24 +
##     pcths25_over + pctbachdeg25_over + pctemployed16_over + pctunemployed16_over +
##     pctprivatecoverage + pctprivatecoveragealone + pctempprivcoverage +
##     pctpubliccoverage + pctpubliccoveragealone + pctwhite + pctblack +
##     pctasian + pctotherrace + pctmarriedhouseholds + birthrate,
##     data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -99.092 -10.939  -0.541   10.684  134.452
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.628e+02  2.592e+01   6.282 4.19e-10 ***
## avganncount    -3.141e-03  9.784e-04  -3.210 0.001351 **
## avgdeathsperyear  1.519e-02  4.668e-03   3.254 0.001160 **
## incidencerate    1.869e-01  9.199e-03  20.313 < 2e-16 ***
## medincome       8.128e-05  1.038e-04   0.783 0.433778
## popest2015     -1.204e-05  6.367e-06  -1.891 0.058845 .
## povertypercent   3.439e-01  2.158e-01   1.594 0.111203
## studypercap     1.234e-03  9.364e-04   1.318 0.187740
## medianage       2.908e-03  1.031e-02   0.282 0.778016
## medianagefemale -1.256e-01  2.837e-01  -0.443 0.658147
## medianagemale   -3.635e-01  2.739e-01  -1.327 0.184616
## percentmarried   1.290e+00  2.206e-01   5.849 5.85e-09 ***
## pctnohs18_24    -3.946e-02  1.480e-01  -0.267 0.789779
## pcths18_24      2.205e-01  1.399e-01   1.576 0.115280
## pctsomecol18_24  6.686e-02  1.434e-01   0.466 0.641108
## pcths25_over    4.235e-01  1.269e-01   3.339 0.000859 ***
## pctbachdeg25_over -1.130e+00  2.012e-01  -5.618 2.24e-08 ***
## pctemployed16_over -7.295e-01  1.471e-01  -4.959 7.74e-07 ***
## pctunemployed16_over  5.654e-02  2.208e-01   0.256 0.797890
## pctprivatecoverage -3.308e-01  2.184e-01  -1.514 0.130113
## pctprivatecoveragealone -7.626e-02  2.411e-01  -0.316 0.751764
## pctempprivcoverage  3.554e-01  1.413e-01   2.515 0.011974 *
## pctpubliccoverage -4.921e-01  3.227e-01  -1.525 0.127451
## pctpubliccoveragealone  6.050e-01  3.818e-01   1.585 0.113216
## pctwhite       -1.351e-01  8.164e-02  -1.654 0.098204 .
## pctblack       -3.383e-02  7.875e-02  -0.430 0.667584
## pctasian        1.473e-01  2.507e-01   0.587 0.556982
## pctotherrace    -7.512e-01  1.594e-01  -4.712 2.64e-06 ***
## pctmarriedhouseholds -1.183e+00  2.090e-01  -5.663 1.72e-08 ***
## birthrate      -8.302e-01  2.596e-01  -3.198 0.001409 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.52 on 1801 degrees of freedom
## Multiple R-squared:  0.5194, Adjusted R-squared:  0.5117
## F-statistic: 67.12 on 29 and 1801 DF,  p-value: < 2.2e-16
```

```
vif(m1)
```

```
##          avganncount          avgdeathsperyear          incidencerate
##          11.594920          34.239132          1.267496
##          medincome          popest2015          povertypersent
##          7.885447          27.602220          9.196313
##          studyperscap          medianage          medianagefemale
##          1.060997          1.028181          10.442018
##          medianagemale          percentmarried          pctnohs18_24
##          9.702684          10.616618          6.598228
##          pcths18_24          pctsomecol18_24          pcths25_over
##          7.558002          11.384527          3.859788
##          pctbachdeg25_over          pctemployed16_over          pctunemployed16_over
##          5.819827          6.927562          2.798019
##          pctprivatecoverage          pctprivatecoveragealone          pctempprivcoverage
##          25.925505          27.039860          8.706406
##          pctpubliccoverage          pctpubliccoveragealone          pctwhite
##          30.715304          25.687141          8.233639
##          pctblack          pctasian          pctotherrace
##          6.222708          2.022906          1.646778
##          pctmarriedhouseholds          birthrate
##          8.727533          1.206980
```

```
par(mfrow=c(2,2)) # Change the panel layout to 2 x 2
plot(m1, id.n=0)
```



```
# Mixed (Categorical + numerical)
m2 <- lm(target_deathrate ~ avganncount + avgdeathsperyear + incidencerate + medincome + popest2015 + p
summary(m2)
```

```
##
```

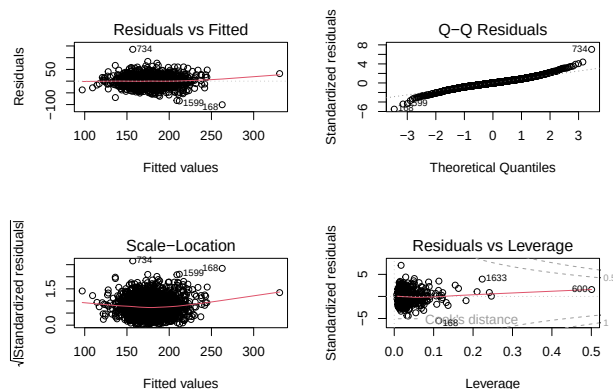
```
## Call:
## lm(formula = target_deathrate ~ avganncount + avgdeathsperyear +
##     incidencerate + medincome + popest2015 + povertypercent +
##     studypercap + medianage + medianagefemale + medianagemale +
##     percentmarried + pctnohs18_24 + pcths18_24 + pctsomecol18_24 +
##     pcths25_over + pctbachdeg25_over + pctemployed16_over + pctunemployed16_over +
##     pctprivatecoverage + pctprivatecoveragealone + pctempprivcoverage +
##     pctpubliccoverage + pctpubliccoveragealone + pctwhite + pctblack +
##     pctasian + pctotherrace + pctmarriedhouseholds + birthrate +
##     binnedinc, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -100.629  -10.876   -0.206   10.737  135.593
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.768e+02  2.649e+01   6.677 3.24e-11 ***
## avganncount      -3.257e-03  9.734e-04  -3.346 0.000836 ***
## avgdeathsperyear  1.695e-02  4.669e-03   3.631 0.000290 ***
## incidencerate     1.883e-01  9.159e-03  20.556 < 2e-16 ***
## medincome         1.252e-04  1.401e-04   0.893 0.371856
## popest2015       -1.340e-05  6.354e-06  -2.108 0.035144 *
## povertypercent    -2.203e-01  2.426e-01  -0.908 0.363998
## studypercap       1.201e-03  9.335e-04   1.287 0.198402
## medianage         2.526e-03  1.027e-02   0.246 0.805771
## medianagefemale   -1.454e-01  2.826e-01  -0.515 0.606882
## medianagemale     -4.269e-01  2.731e-01  -1.563 0.118171
## percentmarried     1.163e+00  2.215e-01   5.250 1.70e-07 ***
## pctnohs18_24       2.169e-02  1.480e-01   0.147 0.883446
## pcths18_24        2.778e-01  1.406e-01   1.976 0.048304 *
## pctsomecol18_24    1.325e-01  1.437e-01   0.922 0.356436
## pcths25_over       3.958e-01  1.271e-01   3.115 0.001867 **
## pctbachdeg25_over  -1.178e+00  2.006e-01  -5.872 5.13e-09 ***
## pctemployed16_over -6.869e-01  1.478e-01  -4.648 3.59e-06 ***
## pctunemployed16_over 4.850e-02  2.199e-01   0.221 0.825444
## pctprivatecoverage -1.827e-01  2.203e-01  -0.830 0.406853
## pctprivatecoveragealone -1.262e-01  2.410e-01  -0.524 0.600574
## pctempprivcoverage  3.099e-01  1.410e-01   2.198 0.028058 *
## pctpubliccoverage  -6.010e-01  3.226e-01  -1.863 0.062629 .
## pctpubliccoveragealone 7.227e-01  3.811e-01   1.896 0.058088 .
## pctwhite          -1.748e-01  8.178e-02  -2.137 0.032742 *
## pctblack          -9.458e-02  7.953e-02  -1.189 0.234512
## pctasian          -6.375e-03  2.541e-01  -0.025 0.979987
## pctotherrace       -7.470e-01  1.594e-01  -4.687 2.98e-06 ***
## pctmarriedhouseholds -1.168e+00  2.083e-01  -5.608 2.36e-08 ***
## birthrate         -8.011e-01  2.588e-01  -3.095 0.002000 **
## binnedinc(37413.8, 40362.7] 1.059e-01  2.212e+00   0.048 0.961818
## binnedinc(40362.7, 42724.4] -4.238e+00  2.323e+00  -1.824 0.068285 .
## binnedinc(42724.4, 45201] -5.169e+00  2.445e+00  -2.114 0.034692 *
## binnedinc(45201, 48021.6] -6.204e+00  2.687e+00  -2.309 0.021078 *
## binnedinc(48021.6, 51046.4] -9.182e+00  2.949e+00  -3.113 0.001879 **
## binnedinc(51046.4, 54545.6] -8.511e+00  3.236e+00  -2.630 0.008613 **
## binnedinc(54545.6, 61494.5] -7.882e+00  3.655e+00  -2.157 0.031174 *
```

```
## binnedinc(61494.5, 125635] -8.889e+00 5.125e+00 -1.734 0.083041 .
## binnedinc[22640, 34218.1] 7.418e+00 2.275e+00 3.261 0.001132 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.4 on 1792 degrees of freedom
## Multiple R-squared:  0.5274, Adjusted R-squared:  0.5174
## F-statistic: 52.62 on 38 and 1792 DF, p-value: < 2.2e-16
```

```
vif(m2)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## avganncount      11.613555 1      3.407867
## avgdeathsperyear 34.650166 1      5.886439
## incidencerate     1.271320 1      1.127528
## medincome        14.540321 1      3.813177
## popest2015       27.814444 1      5.273940
## povertypercent    11.755529 1      3.428634
## studypercap       1.066754 1      1.032838
## medianage         1.031554 1      1.015655
## medianagefemale   10.477811 1      3.236945
## medianagemale      9.760332 1      3.124153
## percentmarried    10.829305 1      3.290791
## pctnohs18_24       6.674629 1      2.583530
## pcths18_24         7.721696 1      2.778794
## pctsomecol18_24    11.561547 1      3.400227
## pcths25_over       3.917575 1      1.979286
## pctbachdeg25_over  5.851099 1      2.418905
## pctemployed16_over 7.073991 1      2.659698
## pctunemployed16_over 2.808626 1      1.675896
## pctprivatecoverage 26.673747 1      5.164663
## pctprivatecoveragealone 27.353747 1      5.230081
## pctempprivcoverage 8.771892 1      2.961738
## pctpubliccoverage 31.048379 1      5.572107
## pctpubliccoveragealone 25.895589 1      5.088771
## pctwhite          8.360438 1      2.891442
## pctblack          6.421660 1      2.534099
## pctasian          2.103002 1      1.450173
## pctotherrace      1.664972 1      1.290338
## pctmarriedhouseholds 8.771906 1      2.961740
## birthrate         1.214102 1      1.101863
## binnedinc         23.824945 9      1.192619
```

```
plot(m2)
```



Following up, we will build m3, a modification of m2 removing some inflating variables (avgdeathspereyear, pctprivatecoveragealone, pctpubliccoveragealone, binnedinc, percentmarried, medianagefemale, medianagemale) which some of them are redundant. This is one approach for addressing multicollinearity in the model, another strategy could be to perform a dimensionality reduction technique like PCA. The results we get from this variable removal is a model with all the independent variables having a less than 10 VIF (However, we still have variables that are above 5 VIF like avganncount, popest2015...), yet still retaining over 50 R-squared. However, the residual analysis results are pretty much the same as in the past 2 models with minor improvements. Anova nested model test confirms that is worth using the lighter model.

```
m3 <- lm(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertyperecent + studyp
vif(m3)
```

```
##          avganncount      incidencerate      medincome
##          9.124716         1.216248         7.126574
##          popest2015      povertyperecent      studyperecap
##          9.218103         8.157984         1.055402
##          medianage      percentmarried      pctnohs18_24
##          1.018945         3.163697         6.397741
##          pcths18_24      pctsomecol18_24      pcths25_over
##          7.398741         10.964065         3.738894
##          pctbachdeg25_over      pctemployed16_over      pctunemployed16_over
##          5.532954         4.725810         2.704761
##          pcttempprivcoverage      pctpubliccoverage      pctprivatecoverage
##          6.285985         4.680138         8.537468
##          pctwhite      pctblack      pctasian
##          7.997588         5.846823         2.006455
##          pctotherrace      birthrate
##          1.630587         1.141253
```

```
summary(m3)
```

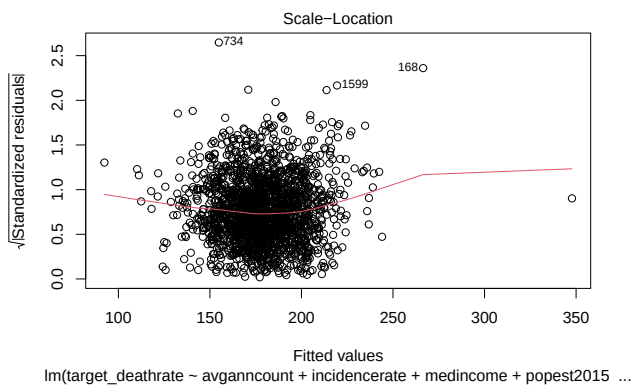
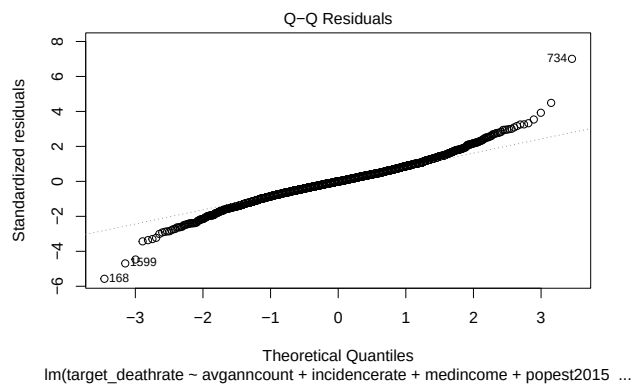
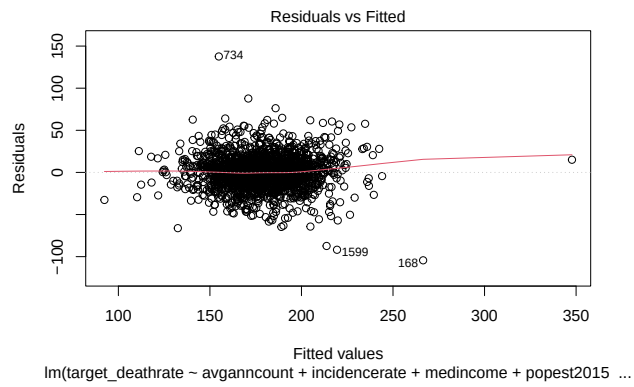
```
##
## Call:
## lm(formula = target_deathrate ~ avganncount + incidencerate +
##      medincome + popest2015 + povertyperecent + studyperecap + medianage +
##      percentmarried + pctnohs18_24 + pcths18_24 + pctsomecol18_24 +
##      pcths25_over + pctbachdeg25_over + pctemployed16_over + pctunemployed16_over +
```

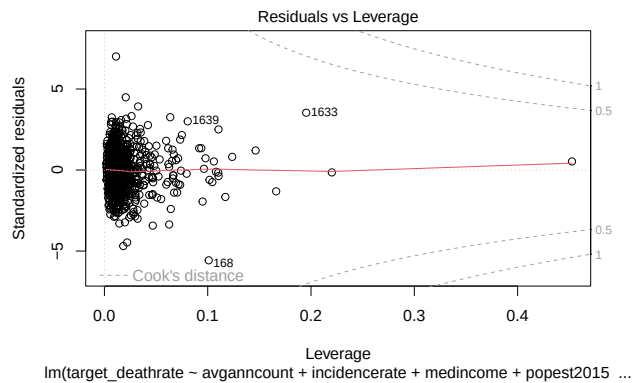
```

##      pctempprivcoverage + pctpubliccoverage + pctprivatecoverage +
##      pctwhite + pctblack + pctasian + pctotherrace + birthrate,
##      data = train)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -104.356  -10.830   -0.097   10.662  137.645
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.350e+02  2.491e+01   5.419 6.81e-08 ***
## avganncount    -1.602e-03  8.784e-04  -1.823 0.068423 .
## incidencerate    1.963e-01  9.120e-03  21.526 < 2e-16 ***
## medincome     -8.366e-05  9.988e-05  -0.838 0.402375
## popest2015     4.595e-06  3.724e-06   1.234 0.217422
## povertypercent  4.592e-01  2.057e-01   2.232 0.025724 *
## studypercap     7.775e-04  9.452e-04   0.823 0.410886
## medianage      2.938e-03  1.039e-02   0.283 0.777415
## percentmarried  1.435e-01  1.219e-01   1.178 0.239004
## pctnohs18_24   -1.776e-01  1.475e-01  -1.204 0.228596
## pcths18_24      1.229e-01  1.401e-01   0.877 0.380712
## pctsomecol18_24 -4.396e-02  1.424e-01  -0.309 0.757616
## pcths25_over     3.712e-01  1.264e-01   2.937 0.003352 **
## pctbachdeg25_over -1.056e+00  1.986e-01  -5.318 1.18e-07 ***
## pctemployed16_over -2.846e-01  1.230e-01  -2.314 0.020758 *
## pctunemployed16_over 2.081e-01  2.197e-01   0.947 0.343652
## pctempprivcoverage 4.319e-01  1.215e-01   3.555 0.000387 ***
## pctpubliccoverage -1.929e-02  1.275e-01  -0.151 0.879772
## pctprivatecoverage -5.798e-01  1.269e-01  -4.571 5.19e-06 ***
## pctwhite       -1.705e-01  8.143e-02  -2.094 0.036363 *
## pctblack       -3.943e-02  7.725e-02  -0.510 0.609839
## pctasian        1.370e-01  2.527e-01   0.542 0.587606
## pctotherrace    -7.710e-01  1.605e-01  -4.803 1.70e-06 ***
## birthrate      -5.905e-01  2.555e-01  -2.311 0.020937 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.75 on 1807 degrees of freedom
## Multiple R-squared:  0.5061, Adjusted R-squared:  0.4998
## F-statistic: 80.5 on 23 and 1807 DF, p-value: < 2.2e-16

```

```
plot(m3)
```





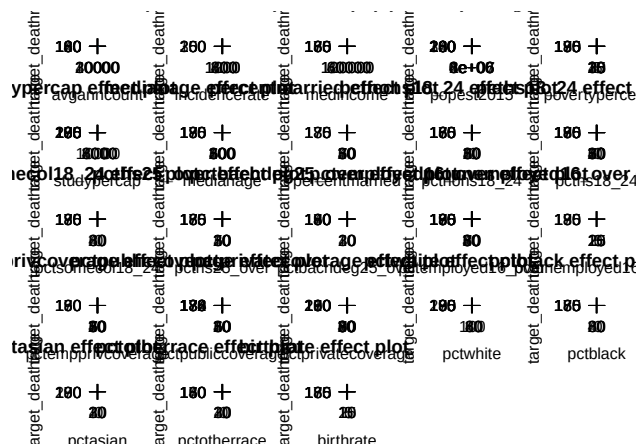
```
#anova(m3, m2)
```

One further improvement for a following model would be to remove variables with high VIF and high p-value in the coefficient tests, like popest2015 or avganncount. Another possible improvement would be to retain binnedinc instead of its redundant counterpart medincome.

```
library(effects)
```

```
## lattice theme set by effectsTheme()
## See ?effectsTheme for details.
```

```
plot(allEffects(m3))
```



Model transformations

Possibly we need to perform some transformations...

```
#Exemple random ;)
#boxcox(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypercent + studypctcap)
#boxTidwell(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypercent + studypctcap)
```

Influential data study

Given that the models are being quite negatively affected by influential data, an study of the aforementioned will be performed to address this issue.

```
#Include Code
```

Further model exploration

```
#Include Code
```

Model validation

In this section we will assess how well the model performs on the training data (train split) to ensure its robustness and capabilities, so we can assess under/over fitting.

We will analyze the residuals to ensure that model assumptions are met and use R-squared as the main performance metric will the model will be computed by means of Ordinary least squares.

```
#Include Code
```

Model evaluation

Finally we will evaluate how well the model performs with unseen data (train split) to assess its forecasting capabilities, using the same R-squared performance metric.

```
#Include Code
```